MPF-III Operation Manual

MPF-III
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Preface

The Multitech Industrial Corporation presented its home computer--MPF-II a couple of years ago. The MPF-II is undoubtedly a success in that it is, among other things, compact and reliable. Earlier this year (1983) we have proudly developed the enhanced version, the MPF-III, a powerful and versatile microcomputer system,to meet the increasing requirement from people engaged in various fields.

This book is written for those who have the MPF-III or want to know about it. We hope you can make use of the system with ease after you have finished reading the book. Below is an overview of what is contained in this manual:

* In chapter 1, "Getting to Know Your MPF-III Personal computer", we give you a general description on the component parts of the system unit as well as an introduction to the peripheral devices and accessories.

* In chapter 2, "Installing the MPF-III", you will learn how to hook up the system.

* In chapter 3,"A Look Inside the MPF-III and a summary of BASIC commands and statements", we present to you the general ideas on how the MPF-III works together with a summary of BASIC commands and statements with which you can write and run BASIC programs on the MPF-III.

* In chapter 4, "More on the Optional devices", a detailed description is provided on the peripheral devices of interests to people in related fields.

* In chapter 5, "Handling the MPF-III with Care", a number of general remarks on care and maintenance of the system is supplied with a view to obtaining best performance of the MPF-III.

* In Appendices, we give you a number of referential informations which you may find useful.

The Multitech Industrial Corporation had tried its best to make this manual informative and instructive. Any suggestions about this manual from the readers for a better revised edition in the future will be appreciated.

Manuals and documents of mutual interests about the MPF-III are expected to be prepared and published in the near future. we are looking forward to receiving opinions and suggestions in this respect from you.

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Chapter 1

GETTING TO KNOW YOUR MPF-III PERSONAL COMPUTER

1.1 Presenting the MPF-III

The MPF-III Personal Computer, an improved version of the MPF-II, is a versatile microcomputer system featuring sophisticated software and hardware capabilities. It is ideal for many dedicated purposes. In this chapter, we will show you the special features of the system itself and the peripheral devices and accessories.

1.1.1 A Look at the MPF-III Personal Computer

As shown in Fig. 1-1, the system unit of MPF-III looks quite different from MPF-II . It takes the shape of a rectangular box. Two disk drives are placed in parallel which can be piled on the system unit as shown in Fig. 1-2. The options such as Z-8Ø CP/M card, floppy disk interface can be installed inside the system unit, which makes the MPF-III a compact, solid, easy-to-transport facility.

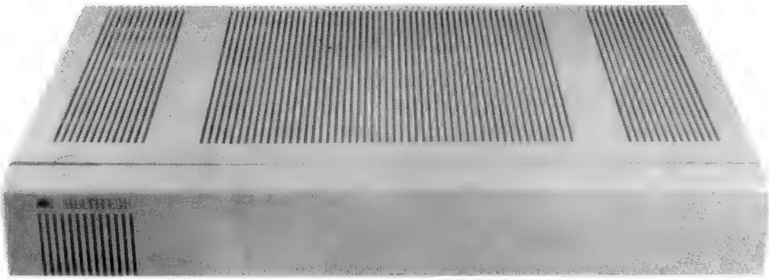


Fig. 1-1 System Unit

FIG. 1-2 The MPF-III Computer System

1.1.2 An Overview of the Specifications

1) CPU

6502 microprocessor

2) ROM

A total of 24K memory capacity, including 11K BASIC interpreter, 4.75K monitor program, 1.25K sound generating program, 3K 80-column text display program and 1.5K printer driver.

3) RAM

A total of 66K memory capacity, including 2K 80-column text display buffer.

4) Screen display:

Text mode : 24 lines with 40 or 80 characters on each line can be displayed.

Low-resolution graphics mode : 40 x 48 blocks can be displayed in any of 16 colors.

High-resolution graphics mode : 280 x 192 dots can be displayed in any of 6 colors.

5) Keyboard

The MPF-III has a separate keyboard of ergonomic design. There are a total of 90 keys on the keyboard. In addition to 56 general-purpose keys, there are

F1--F12 key (User defined key)
PB0--PB1 key
BREAK key
RESET key
LIST key
RUN key
DELC key
INSC key
HALT key
ALT key
Numeric pad

6) Printer Interface and Driver

The MPF-III can be connected to EPSON, C.ITOH, or CP8Ø printers without any modification of the software.

7) Cassette recorder can be connected to the MPF-III as an external source for data storage or retrieval.

8) Display Interface

Either a TV set or video monitor can be used for screen display. (NTSC only)

9) Joystick Interface

The joystick may be used.

1Ø) Sound Generator

With BASIC special sound effect commands you can hear the sound of the machine gun, droping a bomb, explosion, and the raser gun. The sound of the piano, the organ, the bell and the xylophone can be played in the range of three octaves with rhythm accompaniments of waltz, tango, rumba, disco, swing, blues, rock and chacha. The duration of each sound is in the range from that of a whole note to that of a sixty-fourth note.

11) Disk Drive Interface

Disk drives can be connected to the MPF-III.

12) Z-8Ø CP/M card can be added to run CP/M and Microsoft BASIC on the MPF-III.

1.1.3 A List of Standard Optional Devices

1) Disk drive

2) Printer

3) Cassette recorder

4) Remote controller

5) Z-8Ø CP/M card

6) Video monitor(MVM-9G)

7) Serial interface card

8) Floppy disk interface

1.2 Hardware Structure

1) CPU 6502 1MHZ

2) ROM : 24K. There are three 2764 sockets on the system unit.

3) RAM : 64K dynamic RAM and 2K static RAM. The 2K RAM is used for 80-column text display.

4) Screen display

   Text mode 24 lines with 40 or 80 characters on each line.

   Low-resolution graphics mode 40 by 48 blocks in any of 16 colors.

   High-resolution graphics mode 280 by 192 dots in any of 6 colors.

5) Screen : NTSC Composite video, with monitor or TV interface.

6) Keyboard : Easy to use, separate typewriter keyboard with 90 keys.

7) Cassette interface The cassette recorder is connected to store on or retrieve from the magnetic tapes as a secondary memory.

8) Printer interface : With switch select of program control you can choose to connect the MPF-III to an EPSON or C.ITOH printer without changes of any software.

9) Sound Generator : A number of subroutines are available to produce various fundamental sounds.

10) Joystick interface: On the system unit there is a 9-pin D-type connector provided to connect the joystick.

11) System power supply

+5V	4.ØA
+12V	1.5A
-5V	Ø.25A
-12V	Ø.25A

12) Expansion and slot I/O

Slot NO.	Built-in
Ø	16K RAM
1	Printer interface
2	X
3	8Ø-column
4	X
5	X
6	Disk drive interface
7	Z-8Ø CP/M CARD

Slot No. 2 is for external expansion. (compatiable to Apple-IIe bus)

Slot No. 4 is reserved for internal expansion

Slot No. 5 is not available.

1.2.1 Main PC Board

The components configuration of the main PC board is shown in Figure 1.3.

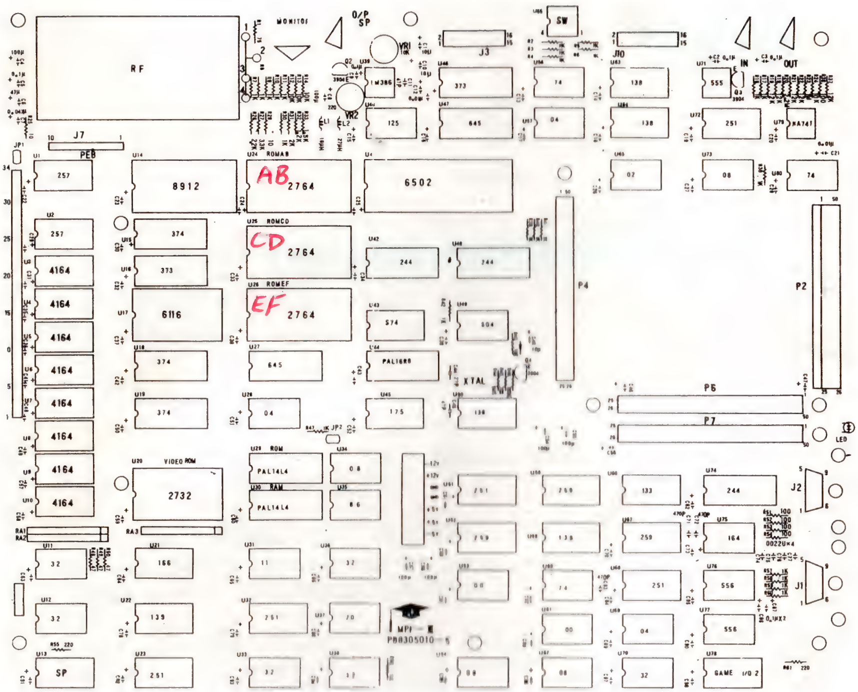


Fig. 1-3 The Main PC Board

This main PC board is the computer itself. On the board there are a total of some 1ØØ IC chips with varied sizes. Among them a big 4Ø-pin IC is the heart of MPF- III, the 65Ø2 microprocessor, which can carry out 5ØØ.ØØØ addition/subtraction operations in a second. The 65Ø2 microprocesser can access 65,536 memory locations. It has a set of 56 instructions and 13 addressing modes. To the right of the 65Ø2 CPU you can find four slots numbered P2, P4, P6, P7 respectively. As a rule, floppy disk interface is inserted in P6, Z-8Ø CP/M card is inserted in P7, P4 is reserved for internal expansion P2 is for external expansion. To the left of slot P4 you can find three ICs numbered 2764, these combine to constitute 24K ROM.

1. Printer interface driver
2. Sound generator
3. 8Ø-column software
4. MPF-III cassette interface software
5. Screen editor
6. BASIC compiler
7. System moniter

To the left of ROM ,you can find an IC numbered 8912, which is a Sound Generator. Six BASIC special commands are available to produce the sound of the piano, the xylophone,the organ and the bell with varied accompaniments in different tempo. In addition, four special sound effects can be produced.

On the left of the main PC board you can find a row of eight ICs numbered 4164 together with a 6116 on the right. They are the RAM of MPF-III, which include 64K dynamic RAM and 2K static RAM for 8Ø-column text display.

These RAM serve as the storage space for your programs and data when you run programs. All the programs and data stored in RAM will disappear if the power supply is turned off.

Through the interfaces constituted by the connectors and components beside the main board, the MPF-III can be connected to printer, disk drive, monitor, cartridge, joystick and keyboard in order to input from or output to the external devices.

On the top of the main PC board you can find a volumn regulator marked VR1 which is used to adjust the volumn of sounds.

To the right of the volumn regulater you can find a row of four miniature switches, the functions of these switches are described below. (You can press them down to get the ON positions.)

S1:on--(See note below)
off--4Ø-column text display

S2:on--Connction to the EPSON or CP-8Ø printer
off--Connection to the C.Itoh printer

S3:on--Channel 3 (NTSC only)
off--Channel 4 (NTSC only)

S4:on--64K RAM is selected
off--48K RAM is selected

Note: The ON state of S1 is reserved for other purposes before you power on the MPF-III, be sure to set this switch to the OFF position. To set the MPF-III in the 8Ø-column mode, use the command PR#3.

1.2.2 Power Supply

A switching power supply is built in MPF-III .The input voltage is 11Ø V AC and the output voltages are +5V +12V -5V and -12V. Below is the pin configuration of the power cord connector.(Fig. 1-4)

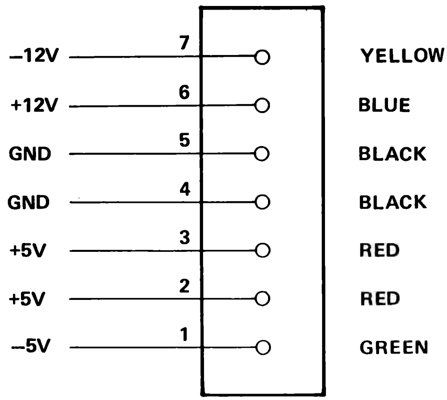


Fig. 1-4 Power Cord Connector

1.2.3 Keyboard

The MPF-III keyboard consists of 90 keys and two LED indicator indicating status of CAPS LOCK and NUM LOCK. The keyboard is connected to the system unit by a cable and a nine-pin D-type connector. The keyboard is of ergonomic design, smart in appearance and ease to operate.

Below is a list of the pin assignments of the MPF-III keyboard cable.

PIN	Signal
1	PB1
2	+5V
3	GND
4	DATA
5	CLOCK
6	STROBE
7	PB0
8	RESET
9	AKD

PBØ, PB1 : Signals read into the system unit. AKD stands for "any key down". If any key is pressed, the signal is high. Otherwise it is low. The codes resulted from a press on the keys are listed in appendix D. All the codes between $8Ø-$FF except $83 and $93 are typamatic (autorepeat) while those between $ØØ-$7F are not. If you hold down a typamatic key for more than Ø.9 second, the code will be sent out repeatedly at a rate of 15 times per second. The codes that can be obtained through the keyboard are $ØØ-$Ø6 and $1F-$FF, totaling 232. ("$" stands for hexadecimal.)

The MPF-III keyboard is shown in FIG 1-5.

Fig. 1-5 The Keyboard

Next we will give you a detailed description of each of the 9Ø keys on the keyboard.

(1) Function keys

On top of the keyboard, you will find 12 function keys numbered F1-F12. In combination of the SHIFT, CTRL and ALT keys, a total of 60 codes are available.

1. Function keys F1-F12 pressed alone --- 12 codes

2. SHIFT key (1) and Function keys pressed simultaneously --- 12 codes

3. CONTROL (CTRL) key and Function keys pressed simultaneously --- 12 codes

4. SHIFT key and CONTROL key pressed simultaneously, followed by a press on the Function keys --- 12 codes

5. ALT key and Function keys pressed simultaneously --- 12 codes

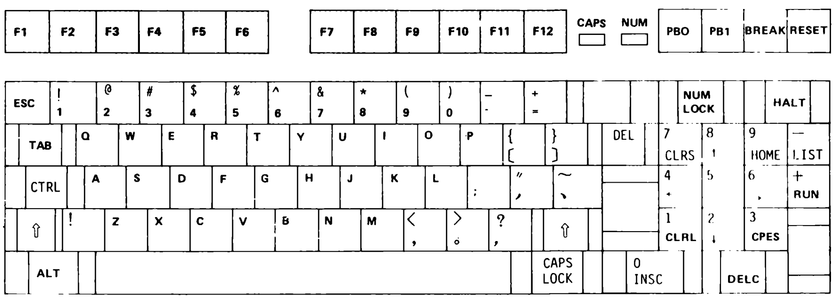


Fig. 1-6 Function key

(2) Special function key

1. PB1: This key is used for self-test of the system. First, press CTRL-RESET, followed by pressing PB1. Release CTRL-RESET first and then PB1. After these steps are completed, the MPF-III will start testing RAM, ROM, together with SG and Color Bar Test.

2. PBØ : If you want to invoke a cold start, you may first press CTRL and RESET simultaneously followed by a press on PBØ and then you release CTRL and RESET followed by PBØ. In so doing you can do without powering off and on the system.

3. BREAK: You may press this key to suspend program execution while you are running a BASIC program. BREAK functions like a CTRL + C combination.

4. RESET: You may press CTRL and RESET simultaneouslly to obtain a warm start.

5. HALT: If you want to suspend the screen display you may press this key. A subsequent press on the same key will resume the screen display. Halt functons like a CTRL + S combination.

6. NUM LOCK : If you press this key, the indicating lamp will light up, and the numeric pad can be used like that found on a desktop calculator. This is a covenient feature when you have a lot of numeric data to handle. A subsequent press on the same key will make the lamp go out and the screen editing function keys are available.

7. CAPS LOCK: On some occasions you may want or need to type only upper case letters. The CAPS LOCK make it easy. Press the CAPS LOCK once to make the lamp light up and all alphabetic characters are displayed in uppercase. No other keys are affected. Press CAPS LOCK a second time the lamp will go out and you have the lower case alphabetic characters.

8. ESC : The ESC key when used in conjunction with I, K, M, J keys move the cursor one postion up, right, down or left. After you enter the 8Ø - column mode with PR#3, ESC 4 can make the system return to 4Ø-column mode and a subsequent ESC 8 will again reenter into the 8Ø-column mode. In 8Ø-column text display mode ESC + R can be used to select upper - or lower case characters. While the CAP lamp is off, characters are entered in lower case. But after pressing ESC R, you may type in upper case letters when the CAPS lamp is off.

In this case, letters enclosed in double quotes remain in lower case.

9. CTRL : CTRL in combination with some other keys can perform a variety of special functions as described earlier.

10. [⇧] : SHIFT key. Holding down this key while pressing another key generates the uppercase letter or the upper character printed on the key.

11. ALT In combination with letter/number keys, this key, this key can perform one-key BASIC command feature.

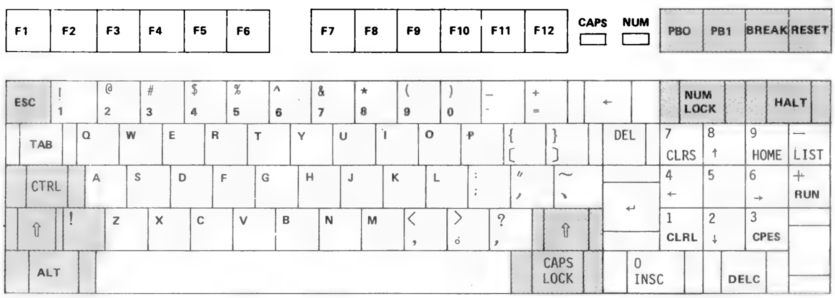


Fig. 1-7 Special Function keys

(3) One-key Screen Editing Commands

To use the one-key screen editing commands, you must key in PR#3 followed by the [↵] key, and then key in ESC and 4 in sequence.

1. INSC : Insert a character. Pressing the INSC key will leave a space where the cursor is positioned, and the character originally at the cursor position as well as those that follow will be shifted one position to the right.

2. DELC : Delete a character. Pressing this key will delete the character currently at the cursor position and the characters that follow are shifted one position to the left.

3. CPES : Copy to end of statement. This key copies from where the cursor is located to the end of the same statement into the buffer. CPES functions the same way you press the [↵] key to the end of the statement.

4. CTRL : Clears from where the cursor is located to the end of the same line.

5. CLRS : Clear all that dislayed on the screen with the cursor remaining at the original position.

6. HOME : This key moves the cursor to the upper lefthand corner with the screen display unaffected.

7. [↓][↑][→][←] : These keys function the same way as M, I, K, J in combination with ESC. In particular, ESC enables/disables this feature. (For more details, see 3.4 screen Editor)

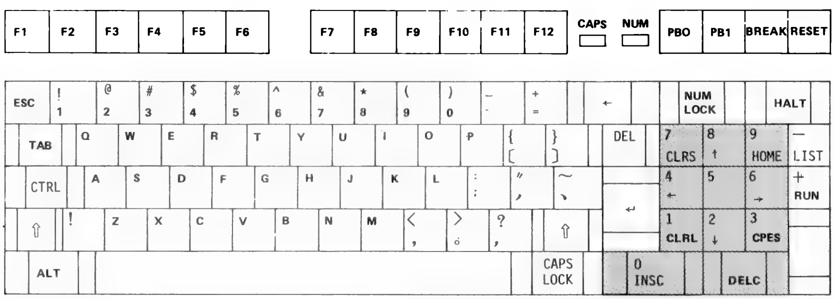


Fig. 1-8 One-key Screen Editing Key

(4) One-Key Command

1. LIST List data and/or program on the video display

2. RUN Run a program

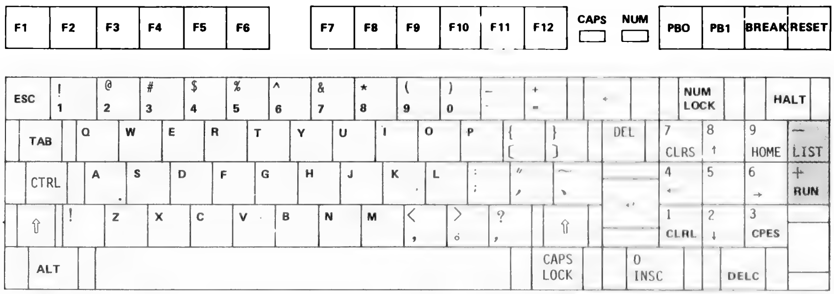


Fig. 1-9 One-Key Command

(5) In the center of the keyboard, you will find 26 letter keys, 1Ø number keys and a total of 3Ø special symbol keys.

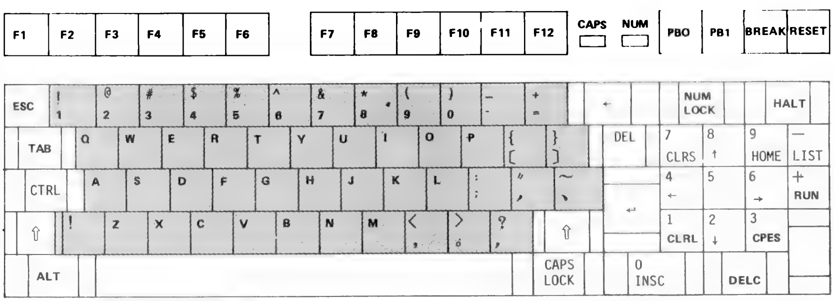


Fig. 1-1Ø The Typewriter Keyboard

(6) Press NUM LOCK key (The lamp lights up) and you have the 10-key numeric pad as found on a calculator for input of numeric data.

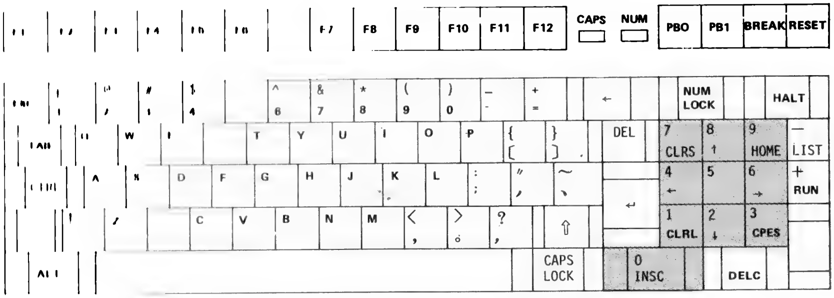


Fig. 1-11 The Numeric Pad

(7) On the bottom of the keyboard you will find a long bar, which is the space key used to produce spaces.

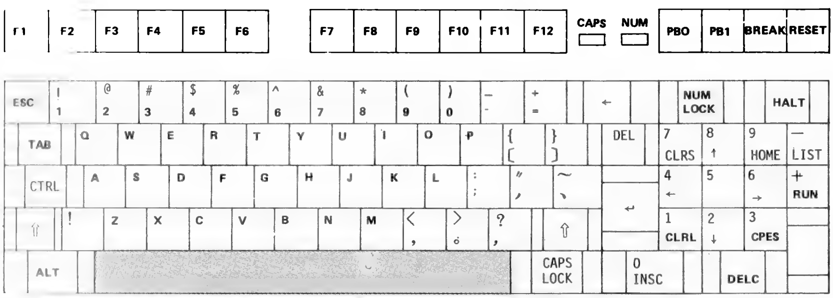


Fig. 1-12 The Space Key

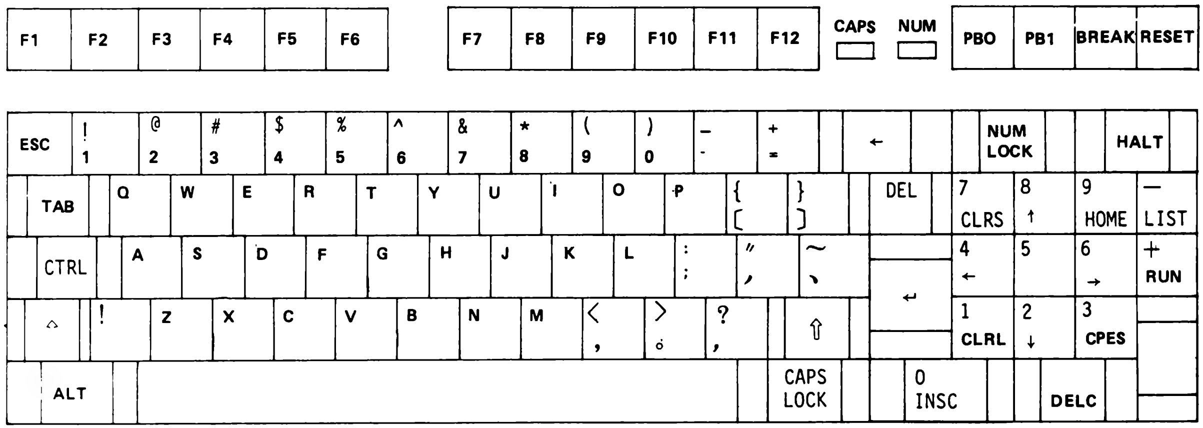


Fig. 1-13 The MPF-III Keyboard

1.2.4 Video Display

On the back panel of the MPF-III system unit, you will find two connectors marked MONITOR and TV respectively. You need a RF modulator if you use TV set instead of a monitor. The TV set must be adjusted to a specified channel.

There are three display modes avàlable on MPF-III as described respectively below.

(1) Text Mode:

In text mode the MPF-III can dislay a screen of 24 lines with 40 or 80 characters on each line. Each character is displayed as a 5 x 7 dot matrix with a one dot space on both sides of the character and above the character.

(2) Low-resolution Graphics Mode:

In low-resolution graphics mode, the MPF-III can dislay an array of 40 by 48 blocks with each block in any of 16 colors available on the MPF-III. As there are no delimiting spaces between blocks in any direction, the adjacent blocks will melt into larger blocks.

(3) High-resolution Graphics Mode:

In high-resolution graphics mode, the MPF-III can dislay an array of 280 by 192 dots in any of six (black, white, orange, blue, green and purple) colors.

In text/low-resolution graphics mode and high-resolution graphics mode, memory locations 0400-0BFF and 2000-5FFF are used as video display buffers respectively. For either mode the source of Information needed to form the screen display are in fact stored in two memory are as of the same size. One is called "primary page" and the other "secondary page". The secondary page is useful when you want an instant display. See the following table:

Mode	Page	Starting Address		Ending Address	
		Hexadecimal	Decimal	Hexadecimal	Decimal
High-resolution	Primary	$2000	8192	$3FFF	16383
Graphics	Secondary	$4000	16384	$5FFF	24575
Text/Low-Resolution	Primary	$0400	1024	$07FF	2047
Graphics	Secondary	$0800	2048	$0BFF	3071

Fig. 1-14 Screen Display Memory Locations

To select which mode and page is to be used for screen display, we use screen "Switches". These switches are controlled by the software (i.e., programs) of the MPF-III. Each switch corresponds to a specific memory location. When a program is used the turn on or turn off (toggle) a switch, it simply references the specific memory location of that switch. The contents of data read from or written to the memory location does not matter at all.

The MPF-III has eight memory locations as soft switches for screen dislay. They come in pairs. In other words, when one of the pair is on, the other is off. The following table lists the addresses of the switches.

Hexadecmal	Decimal	Description
$C050	49232	Display a graphics mode
$C051	49233	Display text mode
$C052	49234	Display all text or graphics
$C053	49235	Mix text and graphics mode
$C054	49236	Display the primary page
$C055	49237	Display the secondary page
$C056	49238	Display low-resolution graphics mode
$C057	49239	Display high-resolution graphics mode

Fig. 1-15 Screen Soft Switches Locations

1.2.5 I/O Expansion Options and Interface Devices

* Printer Interface

On the back panel of MPF-III System Unit, you can see a printer connector marked "PRINTER". The MPF-III recommended EPSON or C.ITOH printers or any other printers with parallel interface the printer driver of which must be able to work well with MPF-III can be cnnected to MPF-III through this connector. The pin configuration of the printer connector is shown in Fig. 1-16.

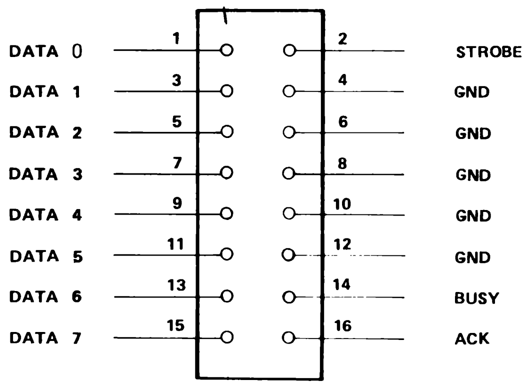


Fig. 1-16 Printer Connector

* Cassette Interface

On the left of the back panel (When holdng an MPF-III with the back panel toward you) of the MPF-III, there are two small holes marked EAR and MIC, respectively. You can plug a cable with a phone plug on each end to your MPF-III and cassette recorder. Then you can store your program from MPF-III to the tape, or load your program from tape to MPF-III.

Note when you load a program from tape to the MPF-III, one end of the cable sould plug to EAR of the MPF-III, while the other end of the cable should connect to ERA or OUT on the cassette recorder. If you store a program from the MPF-III to tape, one end of the cable should plug to MIC of the MPF-III, while the other end of the cable sould cnnect to the jack marked MIC or IN on the cassette recorder.

The connector marked MIC is connected to another soft switch on the MPF-III main board. Like that for the speaker, the soft switch for cassette interface is also a toggle type switch (a two-position switch). The memory location associated with the switch is: 49184 or -16352 (decimal), C020H (hexadecimal). The program that converts data into recordable tone resides in the monitor program.

The other connector marked EAR is for reading program from tape to the MPF-III. Its major purpose is to decode information stored on tape and to load the decoded data into the MPF-III. Thus, a program stored on tape can be loaded back to the MPF-III for execution.

The input circuit of the MPF-III takes 1V peak-to-peak signal from EARPHONE on the recorder and converts the signal to a string of ones and zeroes. Each time the signal received by the input circuit changes from positive to negative or from negative to positive, the state of the input circuit will be changed. If the input circuit was sending ones, it will start sendng zeroes, and vice versa. A program can judge by the value of the memory location 49248 or -16288 or C060H the state of the input circuit. If the value of the location is greater than or equal to 128, then the state of the input circuit is one. Otherwise, the state of the input circuit is zero.

* Speaker Interface

On the left of the MPF-III main board, there is a built-in speaker. You can program the speaker to generate various sounds.

The speaker is controlled by a "soft" switch. The switch can put the paper cone in two positions: "in" and "out". Every time a program references the memory location corresponding to the switch, the state of the speaker is changed. Each time the state of the speaker is changed, the speaker generates a tone. By changing the state of the speaker frequently and continuously, you can generate a tone from the speaker.

The memory location associated with the soft switch is 49200 or -16336 (decimal). The hexadecimal equivalent of that vaue is C030H. When this location is referenced, the speaker will generate a tone.

A program may reference this location by reading from or writing into the location. It is the "referencing" the location that change the state of the speaker. The value that is read from or written into the memory location has nothing to do with the flipping of the soft switch for speaker.

Note that when the 6502 microprocessor performs a "write" operation, it must first perform a "read" operation. Therefore, if you reference the soft switch by writing a value into its associated location, you are actually throwing the switch twice. To a toggle-type switch like the soft switch for the speaker, after a "write" operation is completed, the state of the switch remains unchanged.

1.3 Software Features

In addition to the built-in monitor program, there are certain useful software programs resided in the ROM of your MPF-III. This section will introduce these useful ROM-resident programs to you.

1.3.1 80-column Software

The 80-column program is resided in the ROM of the MPF-III, enabling the MPF-III to have a video display screen of 80 columns (80 characters) wide.

As pointed out in 1.2.1, the S1 of the switch on board location 55 must be set to the off position upon system initialization, then your MPF-III will enter into the 40-column mode and display the cursor "⬟".

When the PR#3 is entered, your video display will enter 80-column mode. To re-enter 40-column mode from 80-column mode, press ESC and [4]. Note that you can not switch to and from 80-column mode without first entering the 80-column mode. ESC [4] causes the display become 40-column mode, ESC [8] switch the terminal from 40-column to 80-column mode.

After entering into 80-column mode, you can use the single keystroke commands and ESC key for screen editing. After the ESC key is pressed, the cursor becomes ⊞ . For detailed desciption of the screen editing features, please refer to Section 3.4.

Control Code	ASCII Name	Name	ASCII Code		Meaning
			Decimal	Hex	
CNT-K	VT	clear EOS	11	ϕB	clear to end of window
CNT-L	FF	home and clear	12	ϕC	move cursor to upper-left cornor of window and clear window
CNT-N	SO	NORMAL	14	ϕE	set display format NORMAL
CNT-O	SI	INVERSE	15	ϕF	set display format INVERSE
CNT-Q	DC1	40-column	17	11	set display to 40 column
CNT-R	DC2	80-column	18	12	set display to 80 column
CNT-S	DC3	stop list	19	13	stop sending data to the display until a key is pressed
CNT-U	NAK	Quit	21	15	1. deactivate 80 column 2. home cursor 3. clear screen 4. reset window
CNT-V	SYN	scroll down	22	16	down scroll, cursor position unchange
CNT-W	ETB	scroll up	23	17	up scroll, cursor position unchange
CNT-Y	EM	home	25	19	move cursor to upper-left cornor of window (but doesn't clear)
CNT-Z	SUB	clear line	26	1A	clear the line the cursor position is ON
CNT-\	FS	forward space	28	1 C	move cursor one space to the right
CNT-]	GS	clear EOL	29	1 D	clear line from cursor position to the end of window

Table 1-1 The Control Characters Available in the 8Ø-column mode

1.3.2 Printer Driver

A ROM-resident printer driver is provided on your MPF-III, enabling your MPF-III to drive various types of printers such as Epson or C.ITOH printers. Various printers are connected to the MPF-III according to the setting of the switch on board location U55 or software.

In addition, printer control messages may be sent to the printer driver to control the functions of the printer. The printer control messages are sent to the printer driver in the form of ASCII codes via the MPF-III BASIC Interpreter. The print control messages are significant to the print driver. However, they are not of any significant to the BASIC Interpreter.

All the printer control messages can be input to the printer driver using software -- simple BASIC commands. However, in some cases, the printer control messages may be input to the printer directly from the keyboard.

1) Select printer using a switch

   If the S2 of the switch on U55 is on, then Epson printer can be connected to your MPF-III. Otherwise, C.ITOH printer is selected.

2) Select printer using software

   * Selecting the C-ITOH printer

   When the ASCII codes 89H and C3H (H stands for hexidecimal. Hexadecimal numbers hereafter in this manual is represented this way.) are sent to the printer driver, the printer driver will select the C.ITOH printer.

   Note that the standard ASCII code for C is 43H. However, the MPF-III ASCII code for C is C3H. Because when the MPF-III BASIC Interpreter processes ASCII codes, the BASIC Interpreter will add 80H to the standard ASCII code of the character involved. Hence, the C3H = 80H + 43H = the ASCII code for the character C.

These two control codes can be input directly from the keyboard by typing CONTROL-I then typing the C key. CONTROL-I is entered by typing the I key while holding down the CONTROL key.

To select the C.ITOH printer by software, enter the following command line:

PRINT CHR$(9);"C"

* Selecting the Epson Printer

When the ASCII codes 89H and 80H+"N" are sent to the printer driver, the printer driver will select the Epson printer.

These two control codes can be input directly from the keyboard by typing CONTROL-I then typing the N key. CONTROL-I is entered by typing the I key while holding down the CONTROL key.

To select the Epson printer by software, enter the following command line:

PRINT CHR$(9);"N"

3) The functioning of the printer can also be controlled by the software. Detailed explanation of how the print controls affect the print effects is presented in a latter chapter. The manuals of Epson or C.ITOH printer should be consulted by those users who would like to use the printer to its full power. A summary of printer controls is presented in this section.

(I) Controlling the Size of Characters to be Printed

The size of letters to be printed by the printers can be adjusted (enlarged or reduced) by sending various ASCII codes to the printer driver. As for which codes are entered for what effects, refer to the paragraphs that follows:

(a) The C.ITOH Printer

Five different sizes of characters can be printed by the C.ITOH printer, including the normal size, enlarged size, and three different sizes which are all smaller than the normal size.

If no ASCII code is entered to alter the print size, then the C.ITOH printer automatically selects a size and prints characters in that size. In this case, the C.ITOH selects the normal size, and we say the printer selects the default size.

To have the printer prints letter in enlarged size. Use the following BASIC statement in your program:

```
PRINT CHR$(14)
```

This statement will send an ASCII code to the printer driver.

If the printer currently prints letters in enlarged size, but the letter size you desire is normal size, then you can enter the following command to tell the printer to switch back to the normal size:

```
PRINT CHR$(15)
```

The three smaller sizes can be selected by entering

ESC, "E" (Input the ASCII codes for ESC and the E.)
ESC, "Q"
ESC, "P"

This can be achieved by entering the statements
^N Pica

"PRINT CHR$(27);"E" Elite
"PRINT CHR$(27);"Q" Compressed (smallest)
"PRINT CHR$(27);"P" Proportional

because once the ESC key is pressed, the MPF-III will enter into the Screen Editor, making it impossible for the printer driver to receive the ASCII code for ESC. Therefore, the ASCII code for ESC is input to the printer driver by software.

To switch back to normal size, enter

ESC, "N"

(b) Epson Printer

The Epson printer prints characters in three sizes __ normal, enlarged, and reduced sizes.

* To print in enlarged size, include the following command in your program:

  PRINT CHR$(14)

However, a carriage return code will cause the PRINT command to print characters in normal size. In other words, a carriage return code will cause the printer to switch back to its default print size.

The print command 'PRINT CHR$(14);"ABCD"' will print ABCD in enlarged size. However, since the above command line is ended with a carriage return, the next print command will print letters in normal size.

* To print in reduced size, include the following command in your program:

  PRINT CHR$(15)

  To switch back to normal size, enter the following command:

  PRINT CHR$(10) or
  PRINT CHR$(18)

(II) To Print with Underline or Emphasized Features

Only the C.ITOH printer offers the Underline and Emphasize features.

* Underline

  To underline a line of printed characters, enter

  ESC, "X"

  To reset the printer so that the Underline feature is suppressed, enter

  ESC, "Y"

* Emphasize

  To emphasize a line of printed characters, enter

  ESC, "!"

  To reset the printer so that the Emphasize feature is suppressed, enter

  ESC,

(Ⅲ) To Change Line Feed Space

The line feed space for both Epson and C.ITOH printers can be adjusted by the software. Currently the line feed space of most popular printers is 1/6 inch. This means that most printers print six lines per inch.

If the line feed space is not specified by the user, the printer driver automatically selects a line feed space and prints accordingly. In this case, we say the default line feed space is selected. The default line feed space set by the MPF-III printer driver is six line per inch.

* Setting the Line Feed Space for C.ITOH Printer

Two easy ways may be used by the user to select line feed space when the user uses the C.ITOH printer. The user may enter the following in your program:

ESC,"A" to select a line feed space of 1/6 inch
ESC,"B" to select a line feed space of 1/8 inch

The line feed space may also be set as you wish by entering the following ASCII codes to the printer driver.

ESC,"T",48+n1,48+n2

to set line feed space to

```
n1*10 + n2
---------- inch
    144
```

The command line

ESC,"T",48+n1,48+n2

is the equivalent of

```
PRINT CHR$(27);"T";"n1";"n2"  or
PRINT CHR$(27);"T";CHR$(48+n1);CHR$(48+n2)
```

* Setting the Line Feed Space for Epson Printer

The line feed space of the Epson printer may be set as you wish by entering the ASCII codes to the printer driver.

1. ESC,"A",CHR$(n)

   This command sets line feed space to

   $$\frac{n}{72} \text{ inch}$$

2. ESC, "O" sets line feed space to 1/8 inch

(IV) To Print Dot Matrix Pattern (Bit Image)

Note that the user should first refer to your printer manual on bit image printing before using the printer to print a bit image or a dot matrix pattern.

When the C.ITOH printer is in use, the following control messages should be sent to the printer driver:

ESC,"S", 48+n1, 48+n2, 48+n3, 48+n4.

Where n1, n2, n3, n4 are used to inform the printer driver of the number of bytes used to built the bit image. The number of bytes used to form a bit image is:

n1*1000 + n2*100 + n3*10 + n4 (bytes)

When the Epson printer is in use, the following control messages should be sent to the printer driver:

ESC,"K",n1,n2

Where n1 and n2 are used to inform the printer driver of the number of bytes used to built the bit image. n1 and n2 are two hexadecimal values. The number of bytes used to form a bit image is:

n2*256 + n1 (bytes)

(V) To Change Line Length

If no printer control message for changing line length is sent to the printer driver, the printer driver automatically set line length to 80 characters per line.

The line length control messages are listed as follows:

89H,80H+30H (This control message sets line length to 10 characters/a line)

89H,80H+31H (This control message sets line length to 20 characters/a line)

89H,80H+3FH (This control message sets line length to 160 characters/a line)

137, 128, ?

(VI) To Perform AND, OR, EOR Logic Operations with High Resolution Graphics Page 1 and Page 2

The result of the logical operations is stored in Page 1 by default. If the user intends to store the result of the logic operations in a user-specified page, please refer to the the last paragraph of this section -- The Selection of Hard Copy Page.

89H,"A" requests the printer driver to perform an AND
89H,"O" requests the printer driver to perform an OR
89H,"E" requests the printer driver to perform an EOR

(VII) To Print Hard Copy

(a) Hard Copy Command:

When the ASCII code "91H" is sent to the printer driver, the printer will produce a normal hard copy for what is displayed on the terminal. The ASCII code may be entered by typing CTRL-Q directly on the keyboard or by entering the following command:

PRINT CHR$(17)

Because 17 (decimal) equals to 11H, which is derived from 91H - 80H.

(b) Inverse Hard Copy:

To produce an inverse hard copy, enter the INVERSE command and then the ASCII codes for 89H and "V" using PRINT CHR$(9);"V"

Another way to produce inverse hard copy is by typing CTRL-I, V directly on the keyboard.

To restore the state of the printer so that normal hard copy can be produced, enter

89H, "R" (PRINT CHR$(9);"R")

Or by typing CTRL-I, R directly on the keyboard.

(c) Enlarge or Reduce the Size of the Hard Copy

To double the size of a hard copy horizontally, enter the ASCII codes 89H and that for "X" or

PRINT CHR$(9);"X"

You may also enter the printer control codes directly from the keyboard by typing CTRL-I,X

To restore, enter

89H,"O" (CTRL-I,O)

To reduce the size of the hard copy, enter

89H,"D1" (CTRL-I,D,1)
89H,"D2" (CTRL-I,D,2)
89H,"D3" (CTRL-I,D,3)

To restore, enter

NORMAL, 89H,"DØH" (CTRL-I,DØ)

The Selection of Hard Copy Page

If no printer control codes are entered to the printer driver, the contents of page 1 will be copied on the printer by default.

```
If Page 1 is to be selected, enter
89H,"P"   (CTRL-I,P)

If Page 2 is to be selected, enter
89H,"S"   (CTRL-I,S)
```

TABLE OF PRINTER CONTROL CODE

FUNCTION	CITOH		EPSON	
	ACTIVATE	RESET	ACTIVATE	RESET
SWITCH SELECT PRINTER	OFF		ON	
SOFTWARE SELECT PRINTER	I^{C}C		I^{C}N	
ENIRGE CHARACTER	OEH	OFH	OFH	SAH, 12H
REDUCE SIZE(1)	ESC, "E"	ESC,	OFH	SAH, 12H
REDUCE CHARACTER SIZE (2)	ESC, "Q"	ESC,		
UNDERLINE	ESC, "X"	ESC,		
EMPHASIZE	ESC,	ESC,		
CHAGE LINEFEED SPACE (DEFAULT $\frac{1}{6}$ inch)	ESC, "T", $48+n_1$, $48+n_2$	LINE FEED SPACE $= \frac{10\times n_1 + n_2}{144}$ inch	ESC, "A"	LINEFEED SPACE $\frac{n}{72}$
CHANGE LINEFEED SPACE ($\frac{1}{6}$ inch)	ESC, "A"		ESC, 0CH	
CHANGE LINEFEED SPACE ($\frac{1}{8}$ inch)	ESC, "B"		ESC,	
PRINT PATTERN	ESC, "S", $48+n_1$, $48+n_2$, $48+n_3$, $48+n_4$ $48+n_3$, $48+n_4$	(DATA $=1000\times n_1$ $+100\times n_2+10\times n_3$ $+n_4$ Bytes)	ESC, n_2	(DATA $256\times n_2$ $+n_1$)
CHANGE LINE LENGTH (DEFAULT 80 CPL) HIGH RESOLUTION LOGICAL OPERATION	89H, 80H+30H~ 89H, 80H+3FH	(10 CPL ~160CPL)	89H, 80H+30H 89H, 80H+3FH	(10 CPL ~160 CPL)
AND	I^{C}A		I^{C}A	
OR	I^{C}O		I^{C}O	
EOR	I^{C}E		I^{C}E	
HARD COPY	CTRL-Q		CTRL Q	
DIVERSE HARD COPY	I^{C}V	I^{C}R		I^{C}R
ENLARGE"X" HARD COPY	I^{C}X	I^{C}O		I^{C}Q
REDUCE HARD COPY (—)	I^{C}D1	I^{C}DO		
REDUCE HARD COPY (=)	I^{C}D2	I^{C}DO		
REDUCE HARD COPY (≡)	I^{C}D3	I^{C}DO		
HARD COPY PAGE 1	I^{C}P	I^{C}DO	I^{C}P	
HARD COPY PAGE 2	I^{C}S		I^{C}S	

1.3.3 Screen Editor

The screen editor makes things easy when you want to revise or add new items to your program or data. in this section we will describe briefly how it works. Additionally, we will take a look on some function keys that can be used for editing purposes.

Cursor Movement Keys

* ESC : Escape key (see below)
* [←] : Left-arrow key
* [→] : Right-arrow key
* [↑] : Up-arrow key
* [↓] : Down-arrow key

Cursor Movement-Two Key Sequences

The character keys A,B,C,and D can move the cursor for one position on the screen if you press ESC before these character keys are pressed.

* ESC A Move the cursor to the right.
* ESC B Move the cursor to the left.
* ESC C Move the cursor down.
* ESC D Move the cursor up.

Another set of keys can move the cursor for a number of positions on the screen. These keys function like the arrow keys if you press ESC to enter into "Movement mode" and then use the following keys.

* K: Move the cursor to the right.
* J: Move the cursor to the left.
* M: Move the cursor down.
* I: Move the cursor up.

A few keys ,when used in combination with ESC, can clear the screen.

* ESC E: All characters from the current cursor position to the end of the display line are cleared. The rest of the screen display is not affected. The cursor will stay at the original position.

* ESC F: All characters form the current cursor position to the end of the screen are cleared. The input data and programs are not affected. The cursor will stay at the original position.

* ESC @: The whole screen will blank out and the cursor will be placed at the upper leftest corner of the screen. The input data and program are not affected.

In 8Ø-column mode, a number of function keys can be used for editing purposes.

* CLRS: Clear the whole screen.
* CLRL: Clear a program line.
* HOME: The cursor is placed at the upper leftest corner of the screen The screen is not cleared.
* CPES: Copy to the end of the display line.
* INSC: Insert a character.
* DELC: Delete a character.
* LIST: Display the program lines.
* RUN : Run a program.

For detailed description and examples, please refer to 3.4 Screen Editor.

1.3.4 One-Key BASIC Commands

On the MPF-III, you can write programs in BASIC. To make things easy, the MPF-III provides the user with a number of one-key BASIC cmmands. (For details on the usage and function of each cmmand, please refer to 3-3.)

To use a one key BASIC command, you press the ALT key followed by a single character to get the commands as shown below:

Character	Command
A	ASC
B	HLIN
C	CALL
D	DATA
E	END
F	FOR
G	GOSUB
H	HOME
I	INPUT
J	GOTO
K	COLOR
L	LEN
M	MID$
N	NEXT
O	ONERR
P	PRINT
Q	SQR
R	RETURN
S	STEP
T	THEN
U	USR
V	VLIN
W	WAIT
X	TEXT
Y	TAB
Z	HGR
1	STOP
2	CONT
3	SAVE
4	LOAD
5	HATB
6	VTAB
7	PEEK
8	POKE
9	PR#
Ø	IN#

1.3.5 Sound Generator(SG)

The MPF-III is equipped with a sound generator which enables the user to produce various music tones as well as special sound effects with six BASIC commands as described below.

1)SONG A,B: Tones covering three octaves as well as five special high/low music scales in conjunction with ten different durations. A sets the pitch and B sets the duration.

2)BASS A: Eight kinds of accompaniment.

3)TEMPO A: A total of 16 tempos in beats/minute.

4)INSTR A: Four musical instruments.

5)PLAY A,B: Auto Play. A sets the starting address of the music score, B sets the accompaniment.

6)EFFECT A: Four special sound effects.

In combination with LOOP, READ, and DATA the preceding six BASIC commands can produce music pieces of specified length. Among them, you can use the TEMPO command to hasten the music to get special effects which can hardly be achieved by manual performance on the music instruments. with SG on the MPF-III, You can write music in BASIC or assembly programs as required in different applications. External speakers can be connected to the system unit to enhance the performance of SG.

1.4 Peripheral Devices and Accessories

1.4.1 Disk Drive

1) Size: 5 1/4 inch.

2) Sectoring: soft sectoring.

3) Recording Side: one - sided.

4) Number of Tracks: 35.

5) Number of Sectors per Track: 16

6) Bytes per sector: 256.

7) Storage: 140KB.

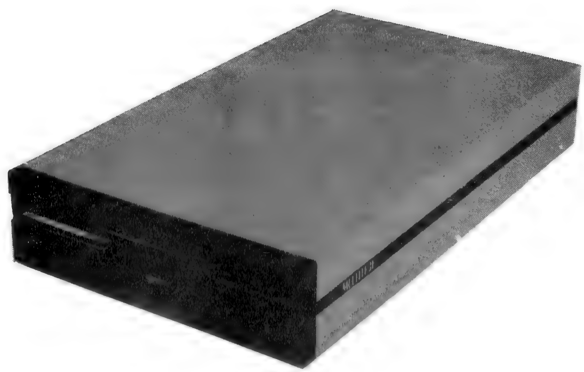


Fig. 1-17 The Disk Drive

1.4.2 Printer

You can display the information or results of the programs on the video screen only temporarily. As you switch off the system unit, the video display will disappear. Therefore, to get a copy of information or results of the programs you need a printer connected to the MPF-III. there are a wide variety of printer suited for different systems amony them, EPSON and C.ITOH printers can be connected to the MPF-III system unit without any interface.

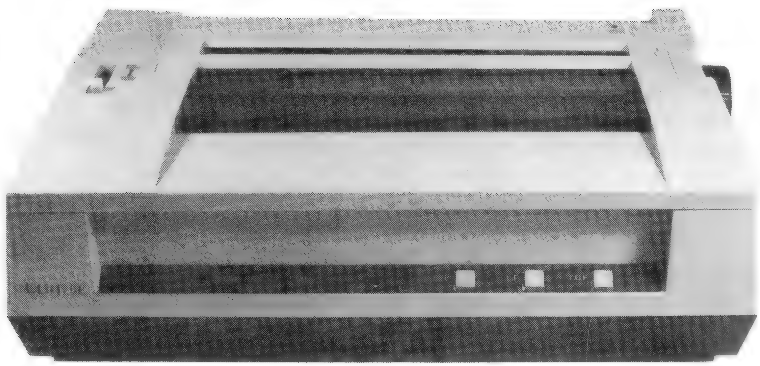


FIG. 1-18 The Printer

1.4.3 Cassette Recorder

The ordinary cassette recorders can be connected to the MPF-III for storage or retrieval of programs and data. The cassette tape is an auxiliary storage and the cassette recorder is an input device as well as well as an output device.

1.4.4 Joystick

The game programs on the tapes or cartridges can be loaded into the MPF-III for you to enjoy TV games. The (the joystick)is used to control the directions of game movements will surely enhance your performance in these games.

1.4.5 Video Display

1) Screen Size: 12-inch

2) Video Frequency Signals:

* Composite Video Frequency Sync.
* Level Ø.5 to 2.Ø v P-P.
* Impedance 75 ohm + High Impedance, Switchable.
* Video Input/Output Connector (RCA Phone Jack).

3) Horizontal Scanning Frequency; 1575Ø ±5ØØ Hz
   Vertical Scanning Frequency: 55 ±1Ø Hz

4) Display Size:
   Width : 2ØØ ±2mm
   Height: 159 ±2mm

5) Controls:
   * Internal: V-Lin, Focus, H-Width
   * Front: Brightness
   * Rear: V-Size
     V-Hold
     CONTRAST
     H-HOLD
     H-PHASE

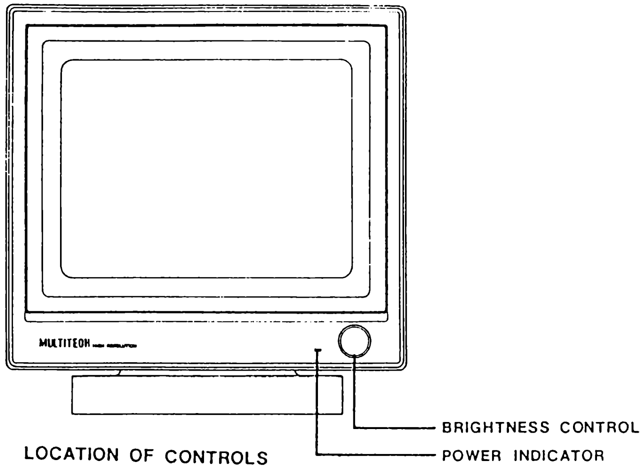


Fig. 1-19 The Video Display (The Front View)

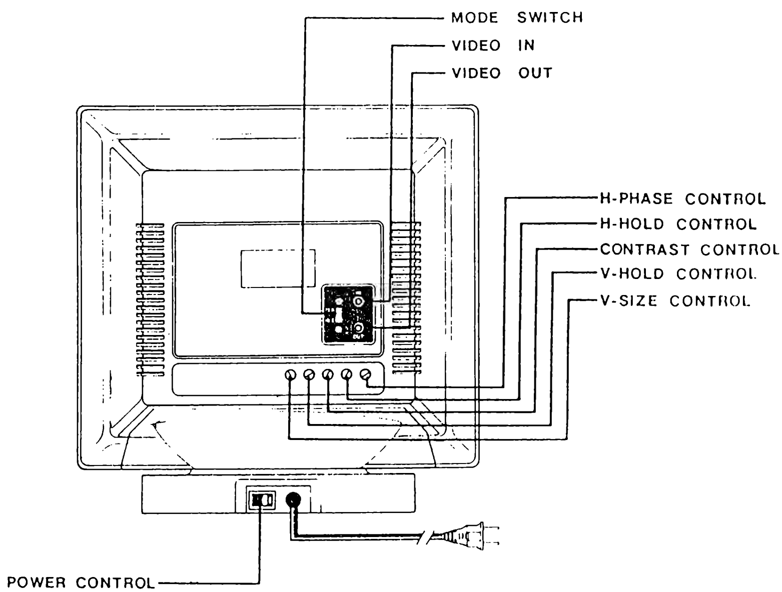


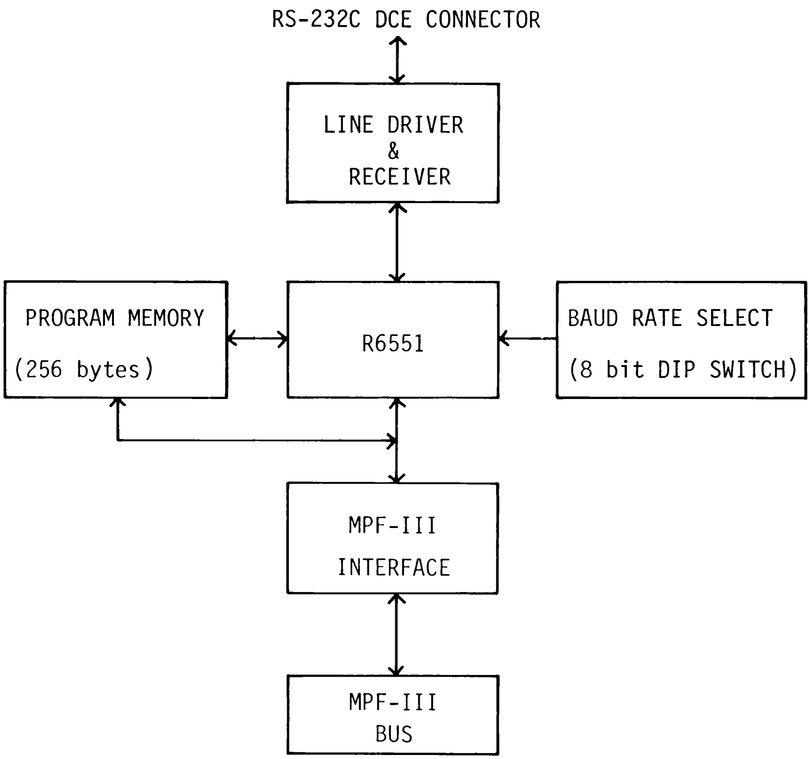
Fig. 1-20 The Video Display (The Back View)

1.4.6 Serial Interface Card

(1) Function

The Serial Interface Card provides the Industry Standard Asynchronous Serial Data Transmission Capability between the MPF-III and external devices.The heart of the SIC is a ACIA (Asynchronous Communication Interface Adapter).

(2) Block Diagram



(3) Specifications

System:
Internal:the MPF-III
External:EIA RS-232-C
Interface Type: DCE(Data Communication Equipment

Data Transmission Mode:
* Asynchronous Serial Data Transmission Mode
* 5-8 data bits
* Odd, even, or no parity bits
* 1, 1½ or 2 stop bits

Data Transmission Rate(clock=16* baud):
Any of the 16 listed below:

External	50 baud	75 baud	110 baud
134.5	150	300	600
1200	1800	2400	3600
4800	7200	9600	19200

Program Memory:

256 bytes ROM

Power Supply

+ 5V D.C.
+12V D.C.
-12V D.C.

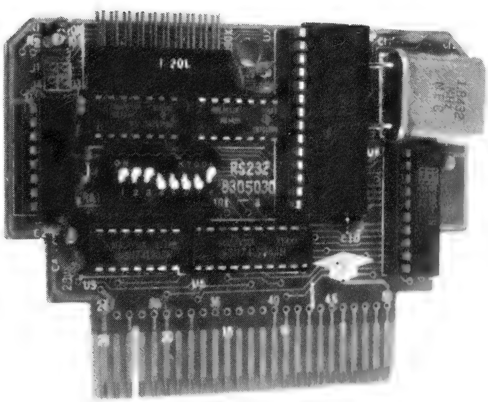


Fig. 1-21 The SIC

1.4.7 Floppy Disk Interface

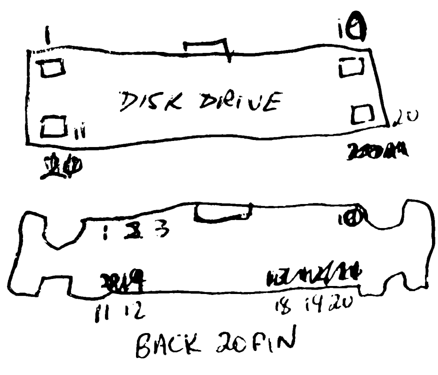
The FDI is the interface between the MPF-III system unit and the disk drive.Two disk drives can be connected to the MPF-III through this interface. Plug the FDI into the specified slot inside the system unit and insert the 20-pin disk drive cables into the connecters marked "DRIVE 1" and "DRIVE 2" on the back panel. Below is a table of the pin configuration of the cable.

PIN	TYPE	PIN	TYPE
1	GND	11	+5V
2	Φ 0	12	+5V
3	GND	13	+12V
4	Φ 1	14	$\overline{\text{ENABLE}}$
5	GND	15	+12V
6	Φ 2	16	READ DATA
7	GND	17	+12V
8	Φ 3	18	WRITE DATA
9	-12V	19	+12V
10	$\overline{\text{WR REQ}}$	20	WRITE PROTECTED

Disk drives with a cable of the same pin configuration may be extected to work well with the MPF-III.

Chapter 2

INSTALLING THE MPF-III



Introduction

This chapter explains how to install the MPF-III and connect the external equipment (peripheral devices). First, please proceed with the following steps:

1) Check the packing list to make sure you have everything you are supposed to have. If necessary, contact your MPF-III dealer.

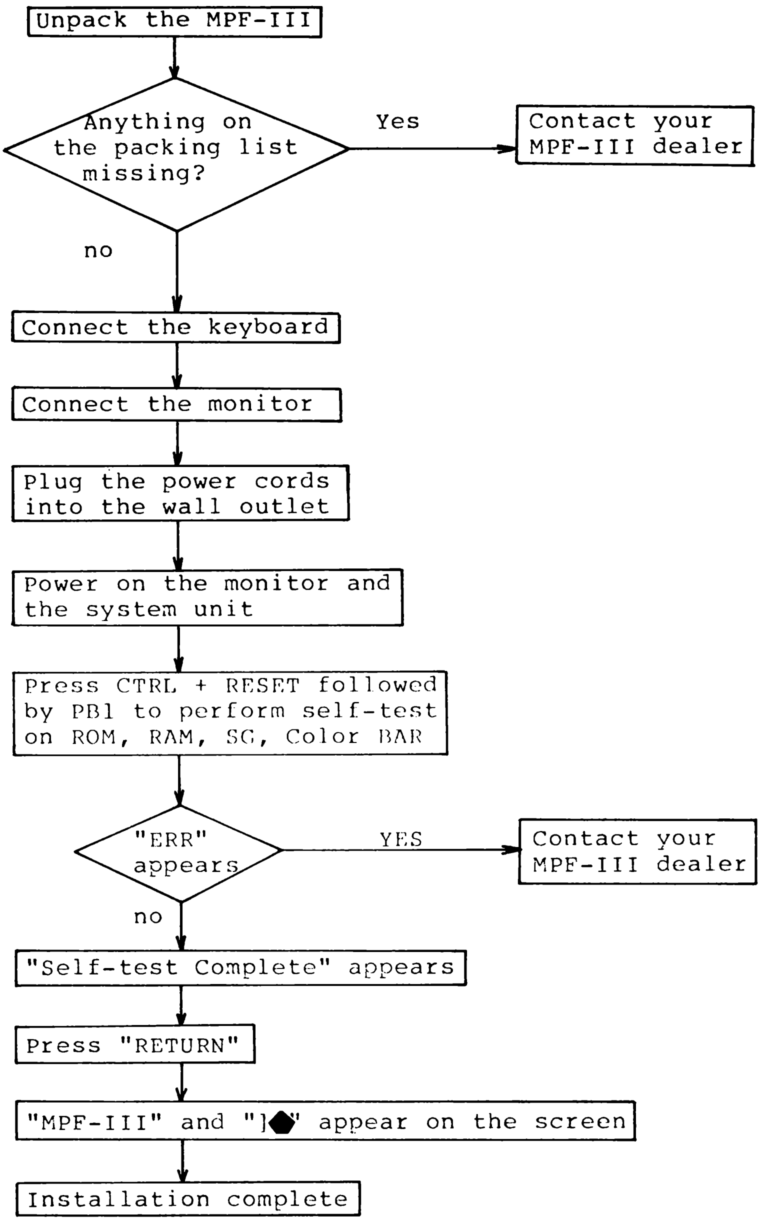
2) Plug the keyboard cable to the connector marked "KEYBOARD" on the right side of the system unit.

3) Connect the monitor cable.

4) Plug the power cords of the MPF-III system unit and the monitor into the wall outlet.

5) Power on the system unit and the monitor and observe the display of "MPF-III" on the monitor.

2.1 Installation Flowchart



2.2 Checking the System Unit and Accessories

1) Check to see if the system unit is firmly equipped. Make sure the case is in good condition.

2) Check to see if the keyboard cable is firmly connected to the keyboard. Make sure the case is defectless.

3) Check the following accessories:

RF modulator
TV cable
Cassette recorder cable
Operation manual
Guarantee card

4) If you find any item stated above missing or in poor conditiion please contact your MPF-III dealer.

2.3 Connecting the Keyboard

1) Plug the cable attached to the upper right of the keyboard into the connector marked "KEYBOARD" on the right of the system unit.

2) Be sure the pins of the cable match those of the connector.

Fig. 2-1 Connecting The Keyboard

2.4 Connecting TV

1) Plug one end of the cable into the connector marked "TV" on the back of the MPF-III and the other end to the side marked "Computer" on the RF modulater (TV/Computer switch box).

2) Disconnect the outdoor VHF antenna from the TV set and connect it to the RF modupator at the side marked "TV".

3) Connect the other two cables from the RF modulator to VHF antenna terminals on the TV set.

4) Switch the RF modulator to the side marked "Computer".When you what to watch TV ,just slide the switch to the "TV" side.

5) If you have a coaxial cable ,disconnect the cable from the TV and screw the cable into the impedance-matching transformer and attach it to the RF modulator.(Consult your local Multitech dealer or distributor, if you don't know exactly how to do.)

6) If you have a video monitor, just connect the cable from the MPF-III to the monitor.

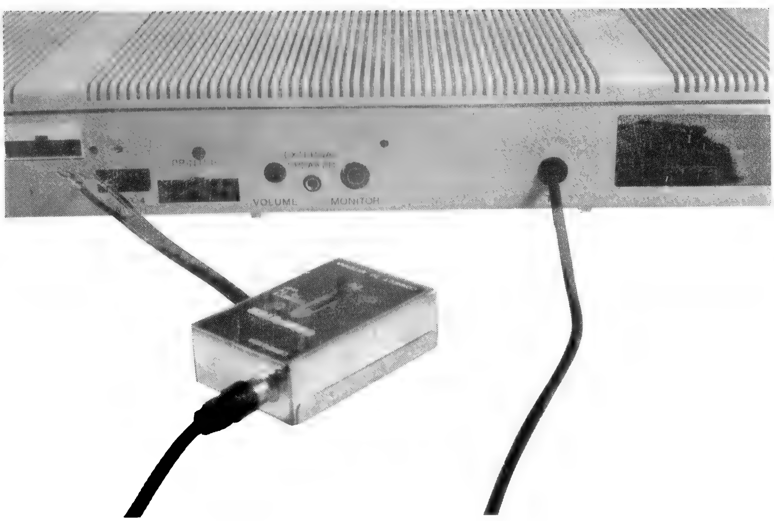


Fig. 2-2 Connecting TV

2.5 Connecting the Disk Drive

(If you order the disk drive together with the system unit, the disk drive and the interface card should have been set ready before delivered to you. In this case, you may skip steps 1 through 4 in the following procedure.)

The disk drive is an optional peripheral device for a wide range of applications. To use a disk drive, you must have a FDI card. Here we will show you how to connect a disk drive.

1) Turn off the power.

2) Remove the screws at the corners of the bottom panel of the system unit to remove the upper cover.

3) Check the components diagram and insert the FDI card into Slot No.6. Be sure you have the pins at the right position and fasten the screws. (Make sure you have the interface card pressed down with a little force to assure good connection.)

4) Replace the upper cover of the system unit and fasten the screws on the bottom panel.

5) Plug the disk drive cable into the connecter marked "DRIVE 1" on the back panel of the system unit. (For a second disk drive use the one marked "DRIVE 2 ".) (When you connect, be sure the side with a small protrusion facing up and the cable fits in the connector squarely with the pins in the right place to avoid accidental damage to disk drive.)

6) Do not turn on the power until all connections of the system are completed. As the power is switched on the disk drive will make whirring and clicking sound for a few moments before "MPF-III" appears on the screen. Soon afterwards the prompt and the cursor will be displayed at the bottom.

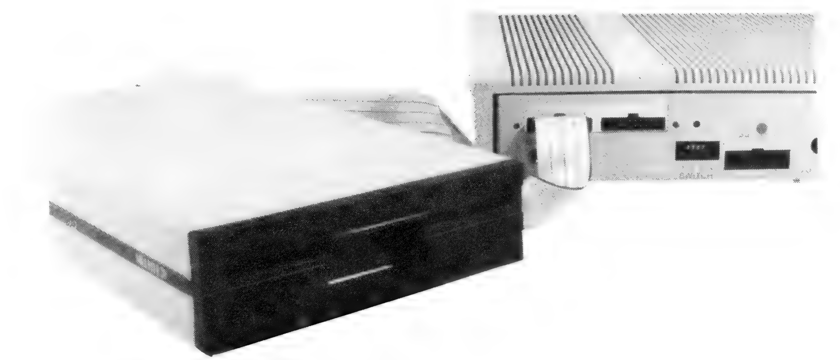


Fig. 2-3 Connecting the Disk Drive

2.6 Connecting the Power Supply

1) Plug one end of the power cord into the connector on the back panel of the system unit and plug the other end into a wall outlet that can supply AC power of correct voltage as shown in Fig. 2-4.

2) Press the switch down (the ON position) and you will see the red power-on LED indicator light up.

3) Press the switch up (OFF position).

4) Turn on the power again. your MPF-III will beep to tell you that it is in good operating condition. Otherwise press the CONTROL and the RESET keys simultaneously and then release. your MPF-III is supposed to beep this time.

5) If your MPF-III doesn't beep any way, there may be somewhere wrong with the connections. Please do the above steps all over again.

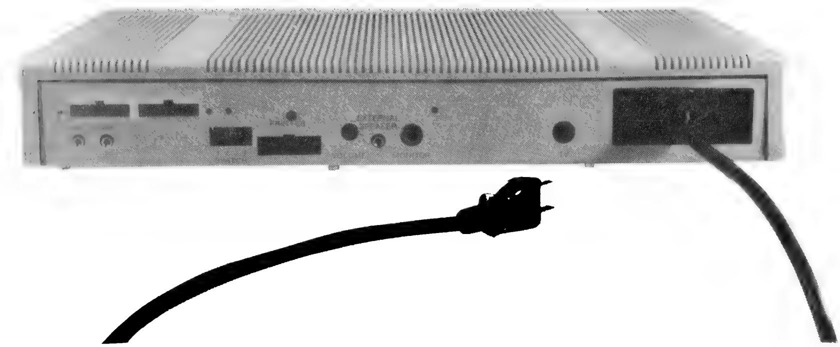


Fig. 2-4 Connecting the Power Supply

2.7 Self-test

There are two ways to perform a self-test:

(1) If you power on the system unit when the keyboard cable is not connected, the system unit will perform a self-test, and if everything is in normal condition, the following messages will be displayed on the screen.

```
* Self-TEST *

(1) RAM (2) ROM (3) SSG (4) COLOR

    RAM     TEST      COMPLETE

    ROM     EF        TEST COMPLETE
    ROM     CD        TEST COMPLETE
    ROM     AB        TEST COMPLETE
```

You will also hear the sound effects. The MPF-III is performing SG Self-test.

You will also see the color stripes displayed.

The MPF-III is performing the color test.

The screen will display:

COLOR TEST COMPLETE
SELF-TEST COMPLETE
PRESS RETURN

Now the self-test is completed.

Connect the keyboard cable and press RETURN, You will see:

]◆

(2) Connect the keyboard cable. press CTRL and RESET **simultaneously followed by PB1. Then release CTRL and RESET first and release PB1 at last. Now the system unit will start self-test. If everything is O.K, you will see the messages as described in (1).**

If there is anything wrong detected during the self-test, you will observe the messages for various errors as follows:

* SELF-TEST *

(1) RAM (2) ROM (3) SSG (4) COLOR

RAM ERR: ^XXXX the address where the error detected is shown after a leading space.

ROM ERR: EF U26 ... Something is wrong with ROM EF at position U26 on the main PC board.

ROM ERR: CD U25 ... Something is wrong with ROM CD at position U25 on the main PC board.

ROM ERR: AB U24 ... Something is wrong with ROM AB at position U24 on the main PC board.

SSG: Special sound effects of bombing, explosion, raser gun and machine gun should be heard if SG is O.K, otherwise there must be something wrong.

COLOR: If the desired colors are not shown, you may adjust the color tuner on the TV set.

If you encounter error messages stated above during the self-test of the system unit, please contact the local dealer of the MPF-III for assistance.

2.8 Connecting the Printer

On the back panel of the MPF-III you will find a row of four tiny switches marked 1, 2, 3, and 4. The switch 2 is used to select which type---EPSON or C.ITOH, will be connected to the MPF-III. When the switch is on, EPSON or CP-8Ø is supposed to be connected, if the switch is off, C.ITOH is assumed.

You can select the printer by software as follows:

1) PRINT CTRL-I, N:for EPSON or CP-8Ø

2) PRINT CTRL-I, C:for C.ITOH

you can do it in a program as follows:

1) PRINT CHR$(9); "C":for C.ITOH

2) PRINT CHR$(9): "N":for EPSON or CP-8Ø

Insert the plug of the printer cable into the connector marked "PRINTER" on the back panel of the MPF-III system unit and set the switch select as described above. With these completed, the MPF-III is ready to print as you request.

Chapter 3

A LOOK INSIDE THE MPF-III AND A SUMMARY OF BASIC COMMANDS AND STATEMENTS

3.1 How the MPF-III Works

Shown in the next page is the block diagram of the structure of the MPF-III system unit, in which you will see:

1) CPU:

   A 65Ø2 CPU with a clock generator

2) MEMORY:

   2K static RAM, 64K dynamic RAM and 24K EPROM

3) VIDEO UNITS:

   Video counter, video latch, video ROM, P/S register (parallel to serial shift register)

4) PERIPHERAL INTERFACES:

   Data latch, SG, S/P register (serial to parallel shift register), RF modulator, flip-flop, MPF-III bus I/O expansion, amplifier.

5) BUS:

   Data bus, address bus, control bus.

How the system works

First, data is input through the keyboard, cassette or exteral sources. It is then processed in S/P register or amplifier and sent to CPU through the buses. It is again processed in CPU and the result is sent to various output interfaces (data latch, SG, RF modulator, flip-flop, etc) and finally sent to the peripheral deveces. The clock generator provides pulses required in operation for CPU and other components. The bus buffers serve as buffers in transmission of data, address and control information. During the data processing in CPU, RAM is used as a workspace for data access, and the programs residing in EPROM control and manage the operations in general. The video units handle the information concerning the video display on the screen.

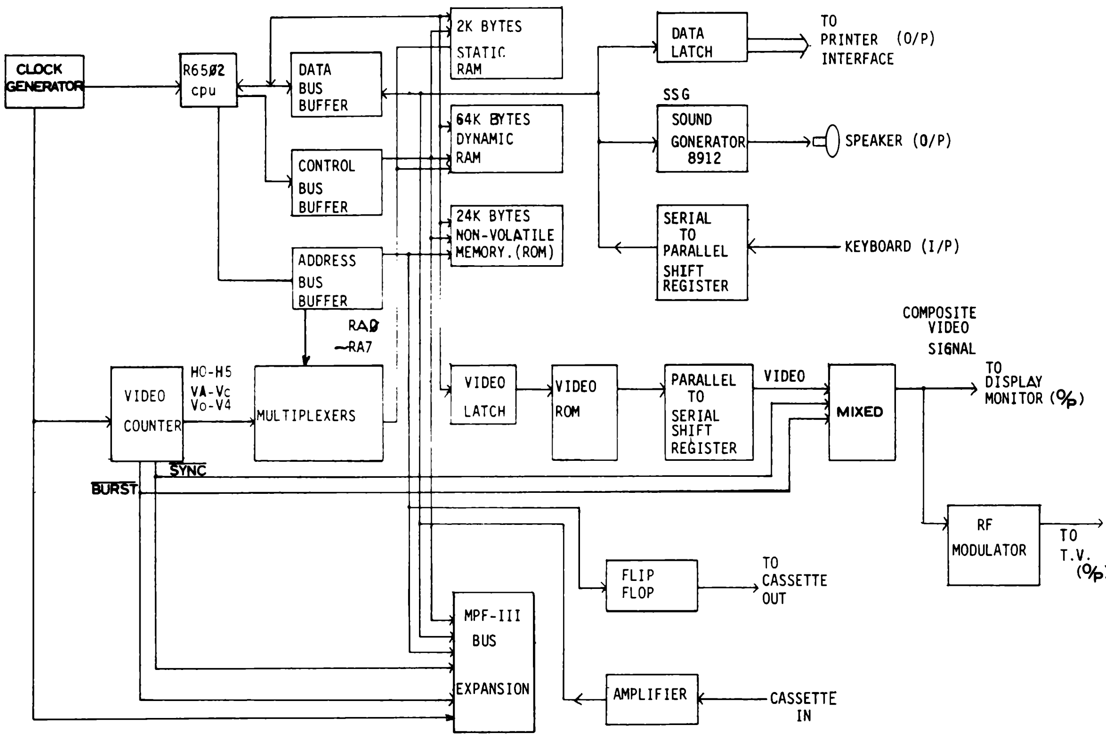


Fig. 3-1 The Block Diagram of the MPF-III System Unit

3.2 Video Display

Below is a general overview on the specifications of MPF-III video display.

1) Display modes:

   Text mode
   Low-resolution graphics mode
   High-resolution graphics mode

2) Text display format:

   24 rows with each row of 40 or 80 characters.

3) Character font:

   5 by 7 dot matrix

4) Character set:

   96 ASCII characters

5) Graphics capability:

   1920 (40 by 48) blocks in low-resolution graphics mode and 53760(280 by 192) dots in high-resolution graphics mode.

6) Character display mode:

   Normal, inverse.

7) Number of colors:

   16 in low- and 6 in high-resolution graphics mode.

3.2.1. Text Mode

In text mode, the MPF-III can display a screen of 24 lines with 40 or 80 characters on each line. Each character on the screen represents the contents of one memory locatiion in the current video display buffer. a total of 90 characters can be displayed on the screen. The character set includes 26 uppercase letters of the alphabet, 26 lowercase letters of the alphabet, 10 numbers and 28 special symbols.

Each character is displayed as a 5 by 7 dot matrix with a one-dot space on both sides of the character and above the character to keep the words apart.

The memory locations used in the text mode is:

$Ø4ØØ-$Ø7FF (primary page)
$Ø8ØØ-$ØBFF (secondary page)

Fig. 3-2 shows the 9Ø characters that can be displayed on the screen.

```
ABCEDFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
[\]-!"#$%&'()*+-./,:;<=>?{}
Ø123456789
```

Fig. 3-2

3.2.2 Low-resolution Graphics

In low resolution graphics mode, the screen can display an array of 4Ø by 48 blocks, while each block may come in any of the 16 colors available on the MPF-III. But the blocks come in only white and grey on monochrome monitors.

Decimal	Hex	Color	Decimal	Hex	Color
Ø	$Ø	Black	8	$8	Brown
1	$1	Magenta	9	$9	Orange
2	$2	Dark blue	1Ø	$A	Grey
3	$3	Crimson	11	$B	Pink
4	$4	Dark Green	12	$C	Light Green
5	$5	Grey	13	$D	Yellow
6	$6	Medium Blue	14	$E	Aquamarine
7	$7	Light blue	15	$F	White

Fig. 3-3 Low-resolution Graphics Colors

3.2.3 High-resolution Graphics

When your MPF-III operates in high resolution graphics mode, the screen displays an array of 53,760 dots (280 by 192). Each dot may come in one of the six colors-- white, black, green, blue, orange, purple.

When operating in high resolution graphics mode, the MPF-III fetches screen information from a memory area consisting of 8, 192 bytes. This memory area is known as display buffer. The display buffer is divided into two areas; page 1 and page 2. The memory range of page 1 or primary page is from 2000H (hexadecimal) or 8192 (decimal) through 3FFFH or 16383, and the memory range of page 2 is from 4000H through 5FFFH.

Each dot displayed on the screen (when the MPF-III operates in high resolution graphics mode) represents one bit in the screen buffer. The seven bits of a byte is displayed on the screen, while the remaining bit is used for selecting color for the seven bits displayed on the screen.

Each line on the screen requires 40 bytes of information. The first bit of the first byte is displayed as the leftmost bit (or at the leftmost position of a line), followed by the second bit, and third bit through the seventh bit. The eighth bit of the first byte is used for selecting color, it is not displayed . Following the seventh bit of the first byte is the first bit of the second byte, and the first bit of the third byte follows the seventh bit of the second byte. Therefore, each line consists of

(dots) x 40 (bytes) = 280 (dots)

On a black-and-white TV or video monitor, if the corresponding bit of a dot is 1 or "on", the dot appears in white on the screen; if the corresponding bit is 0 or "off", the dot comes in black on the screen.

However, on a color TV or monitor, the color of a dot is determined not only by whether its corresponding bit is on or off, but also by the position where the dot appears on the screen. If the corresponding bit of the dot is off, the dot will come in black. But if the corresponding bit of the dot is on, the color of the dot will be decided by its position on a screen.

If the dot appears on the column Ø, leftmost column, or on even-numbered column it will appear violet. If the dot is in the right column (column 279) or an odd-numbered column, it will appear in green. If two dots come side by side, they will be in white. If the dot which is contained in a byte with the eight bit being one, then the purple and green colors will be replaced by blue and orange colors.

In high resolution graphics mode, there are six colors available. However, they are subject to the following limitations.

1) Dots on even-numbered columns only come in black, blue, or purple.

2) Dots on odd-numbered columns only come in black, green, or purple.

3) Each byte must come in either purple/green or blue/orange. It is absolutely impossible that purple/orange, purple/blue, green/orange, green/blue come in the same byte.

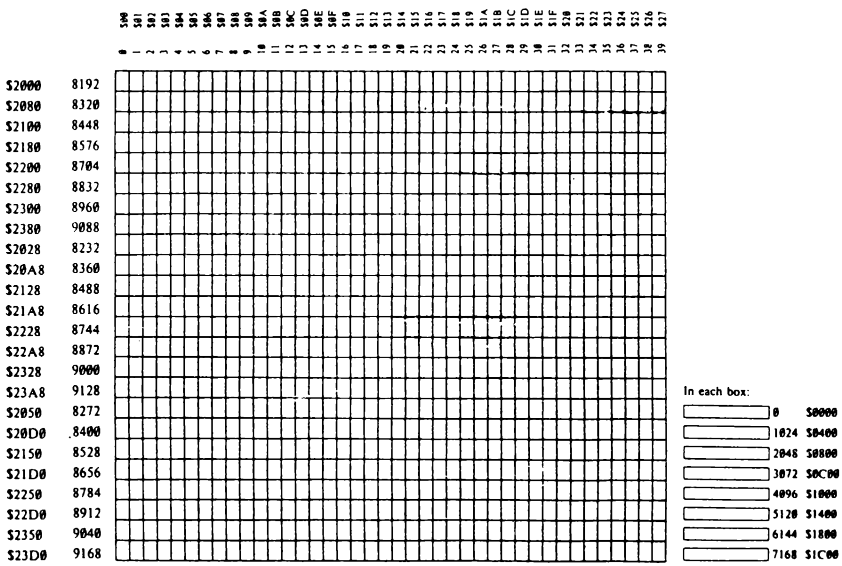


Fig. 3-4 Map of the High Resolution Graphics Mode

And lastly, a few words on the 80-column text display. The MPF-III is built with 80-column text display software and hardware. The program pertaining to the 80-column text display provides the user with a wide variety of pointer control and editing features which will operate with the right selection of the switch. The foundation for 80-column text display is to double the video output in the same period of processing time required in 40-column text display circuitry. To do this, auxiliary memory is required in acddition to the main memory. The display circuitry access the main and the auxiliary memory simultaneously in 80-column text display to double the processing speed.

3.3 BASIC Commands and Statements

BASIC is built into your MPF-III personal computer. As you switch on the computer and see the prompt "]" on the video display, the MPF-III is already in BASIC mode and you can key in BASIC instructions.

(To enter into the monitor, just type CALL-151 [⏎] and you will see the system prompt "*" which tells you that the MPF-III is now in monitor. Then if you want to reenter BASIC, just type CONTROL + C [⏎] .)

In this section, we will present to you a rough description of the BASIC commands and statements. A list of terms with their definitions used in this section is given here.

imm: Immediate execution mode

def: Defered exection mode

| Delimiter of alternatives

{ }: Items that may be repeated

[ ]: Items that are optional

/ /: Value of items enclosed

var: Any variable

avar: Arithmetic variables, name or name%

name: Variable names

name%: Integer variable names

name$: String variable names

aop Arithmetic operators (+,-,*,/)

aexpr: Arithmetic expressions

sexpr: String expressions

literal: String constants composed of characters

character: Alphabets, numbers and special symbols

string: Strings composed of characters enclosed in a pair of quotes.

digit Numbers Ø through 9

sop String operators

expr Numeric and string expressions

linenum Line numbers

reset The RESET key

return The ↵ key

ctrl The CONTROL key

3.3.1. Commands and Statements Relating to System

LOAD and SAVE

Execution mode imm & def

```
Format    LOAD [filename]
          SAVE [filename]
```

The LOAD instruction loads a program on the tape into the MPF-III. To load a program, first type the LOAD instruction and then press PLAY on the cassette recorder. Finally press ↵.

The SAVE instruction stores a program from the MPF-III memory onto the tape. Connect the tape recorder to the MPF-III. Type the SAVE instruction. Press PLAY and RECORD on the cassette recorder. Finally press ↵.

NEW

Execution mode imm & def

Format NEW

The NEW command has no parameters. You can use NEW to completely delete your program and all variables currently in the RAM. This command is normally used just before typing in a new program to get rid of the current program.

RUN

Execution mode imm & def

Format RUN [Linenum]

When you type RUN, the computer will start executing your program at the line number indicated by Linenum and will clear all variables, pointers, and stacks. If you do specify a line number from which you want to start program execution, RUN will start at the lowest numbered line in the program.

Example

```
10 PRINT 4 + 5
20 PRINT 6 + 4
30 PRINT 7 + 6

] RUN 20
10
13
```

STOP, END CTRL-C, CTRL-RESET, CONT

Execution mode   imm & def

Format

```
STOP
END
CONT
```

Execution mode   imm

Format

```
CTRL-C
CTRL-RESET
```

The STOP command can be used to stop program execution, and return control to the user. When you issue a STOP command the following message will be displayed:

BREAK IN XXX (XXX is the line number)

Exemple

```
10 A=5
20 B=6
30 PRINT A+B
40 STOP
]  RUN
11
BREAK IN 40
```

Like STOP, the END command will cause program execution to cease, but no message will be displayed.

Example

```
10 A=5
20 B=6
30 PRINT A+B
40 END
] RUN
11
```

The effect of CTRL-C (Press C while holding down CTRL) is the same as STOP inserted immediately after the statement presently being run.

By using CTRL-RESET ,which has the effect of powering on the MPF-III, you can immediately stop the execution of a program or command. Although you will not lose your program , some pointers or stacks may be cleared upon the execution of CTRL-RESET.
If STOP, END, or CTRL-C is used to cease program execution, the CONT command can be used to instruct the MPF-III to resume execution from the next statement.

TRACE and NOTRACE

Executin mode imm & def

Format TRACE
NOTRACE

The TRACE command can be used to set the computer to a debug mode which will display the line number of each statement executed. The command NOTRACE, the opposite of TRACE, turns off the TRACE debug mode. (Note: You can't end the TRACE session by RUN, CLEAR, NEW, DEL or CRTL-RESET.)

Exmaple

```
10 TRACE
20 FOR I=1 TO 3
30 PRINT I
40 NEXT I
50 NOTRACE
] RUN

#20 #30  1
#40 #30  2
#40 #30  3
#40 #50
```

PEEK

Execution mode imm & def

Format PEEK (aexpr)

PEEK can be used to return the contents, expressed in decimal, of the byte at address /aexpr/.

```
Example    10 A=PEEK(200)
           20 PRINT A
           ] RUN
```

POKE

Execution mode   imm & def

Format   POKE aexpr 1, aexpr 2

POKE can be used to store the binary equivalent of /aexpr 2/ into /aexpr 1/ in memory. The range of /aexpr2/ has to be from 0 through 255, and /aexpr1/ must be from -65535 through 65535.

```
Example    10 A=PEEK(200)
           20 POKE 8000,A
```

WAIT

Execution mode   imm & def

Format   WAIT aexpr 1, aexpr 2 [,aexpr3]

The WAIT command permits the user to place a conditional pause into a program. In Format, the parameter /aexpr 1/ is the address of a memory location which has to be in the range -65535 through 65535. The other two parameters, /aexpr 2/ and /aexpr 3/ used in this command, are decimals which are required to be in the range 0 through 255. The bit pattern of /aexpr 1/ in conjunction with /aexpr 2/ and /aexpr 3/ will decide whether or not the program should be suspended.

Example   WAIT aexpr 1, 255, 0, (The program will halt unless one of the 8 bits is 1.)

WAIT aexpr 1, 3, 2, (The program will not resume until the rightest bit is 1 and the rightest bit but one is 1.)

CALL

Execution mode   imm & def

Format   CALL aexpr

CALL will cause the computer to execute a machine language subroutine whose memory location is specified by the argument /aexpr/.

Example CALL-936

HIMEM:

Execution mode imm & def

Format HIMEM aexpr

The HIMEM: command established the address of the highest memory location available to a BASIC program, including variables.

Example HIMEM: 64000

LOMEM:

Execution mode imm & def

Format LOMEM aexpr

The LOMEM: command sets the address of the lowest memory location available to a BASIC program.

Example LOMEM 50000

USR

Execution mode imm & def

Format USR (aexpr)

USR passes the argument /aexpr/ to a machine language routine.

Example

```
] CALL-151
* OA: 4C 00 03
* 0300:60
] PRINT USR (8)*3
  24
```

3.3.2 Commands and Statements Relating to Format and Editing

LIST

Execution mode   imm & def

Format   LIST [linenum 1] [- linenum 2]
          LIST [linenum 1] [,linenum 2]

LIST is used to display a portion or the whole of a program on the screen.

Example   LIST 10, 30 (Program lines 10 through 30 will be displayed.)

Example   LIST (The whole program will be displayed.)

DEL

Execution mode   imm & def

Format : DEL linenum1[,linenum2]
The DEL command will delete the lines from linenum1 through linenum2 inclusive.

Example   DEL 20, 60

REM

Execution mode   imm & def

Format   REM {character|"}

The REM statement allows text, including blanks and statement delimiter(:),to be inserted in a program; even though whatever typed after REM is ignored by the computer.

Example

```
10 REM COUNT A+B
20 INPUT A
30 INPUT B
40 PRINT A;"+";B;"=";A+B
50 GOTO 20
```

VTAB

Execution mode imm & def

Format VTAB aexpr

VTAB is used to move the cursor vertically to place it in the /aexpr/-th line.

Example
```
10 VTAB20
20 PRINT "MPF"
] RUN
MPF (Printed in the 20th line counted from
the top of the screen.)
```

HTAB

Execution mode imm & def

Format HTAB aexpr

HTAB moves the cursor left or right to the /aexpr/-th position on the line.

Example
```
10 HTAB 15
20 PRINT "MULTITECH"
] RUN
                MULTITECH

      (Printed from the 15th position)
```

TAB

Execution mode imm & def

Format TAB

TAB must be used together with a PRINT statement, and aexpr is required to be enclosed in parentheses. TAB moves the cursor to the position that is /aexpr/ printing positions from the left margin of the text window.

Example
```
10 PRINT TAB(20); "MPF"
] RUN
MPF (Printed from the 20th position.)
```

POS

Execution mode imm & def

Format POS (aexpr)

POS returns the current position of the cursor. The aexpr in the Format is a dummy argument which can be a number, variable, or a string.

Example
```
10 PRINT TAB (19); POS (0)
] RUN
18
```

SPC

Execution mode imm &def

Format SPC (aexpr)

SPC is used to leave /aexpr/ spaces when the result of the program execution is displayed. SPC must be used together with the PRINT statement.

Example
```
10 PRINT "MPF"; SPC (5); "COMPUTER"
] RUN
MPF     COMPUTER
```

HOME

Execution mode imm &def

Format HOME

HOME moves the cursor to the upper leftest corner on the screen and clears the whole screen. The program, variable, and data are not affected.

Example HOME

CLEAR

Execution mode imm & def

Format CLEAR

CLEAR is used to clear all variables, arrays, strings and reset pointers and stacks.

```
Example    10 A = 23
           20 B% = 9
           30 C$ ="AR"
           40 CLEAR
           50 PRINT A;  B%;C$
           ] RUN
           00
```

FRE

Execution mode   imm & def

Format   FRE (expr)

FRE is used to return the amount of memory positions available. The argument expr is a dummy argument which can be any number, variable, or string.

```
Example    10 PRINT FRE(0)
           ] RUN
```

INVERSE and NORMAL

Execution mode   imm & def

Format   INVERSE
NORMAL

INVERSE sets the video mode so that what was black on the screen is changed to white and what was white is changed to black. White characters originally on a balck background is changed to black characters on a white background.

NORMAL sets the video mode so that the usual white characters appear on a black background.

SPEED

Execution mode: imm & def

Format   SPEED = aexpr

The SPEED command regulates the speed at which characters are to be returned to the screen or sent to other input/output devices. The value of aexpr must be in the range 0 through 255.

```
Example    10 SPEED = 0
           20 PRINT "MULTITECH"
           ] RUN
           MULITITECH
```

ESC $\boxed{\uparrow}$

Executon mode   imm (for editing purposes)

Format   ESC $\boxed{\uparrow}$

Press ESC once and subsequent press on the $\boxed{\uparrow}$ key will move the cursor upward. A second press on ECS will disable the cursor movement feature of (ESC is a toggle switch.)

$\boxed{\rightarrow}$ $\boxed{\leftarrow}$ $\boxed{\downarrow}$

Execution mode: imm

Format: $\boxed{\rightarrow}$ $\boxed{\leftarrow}$ $\boxed{\downarrow}$

$\boxed{\rightarrow}$ $\boxed{\leftarrow}$ $\boxed{\downarrow}$ moves the cursor right, left and down respectively.

CTRL-X

Execution mode   imm

Format   CTRL-X

CTRL-X tells MPF-III to ignore the instruction on the current input line.

3.3.3 Commands and Statements Relating to Arrays and Strings

ASC

Execution mode   imm & def

Format ASC (sexpr)

ASC returns the ASCII code of the first character of the string sexpr.

```
Example    10 PRINT ASC ("MPF")
           ] RUN
           77
```

CHR$

Execution mode imm & def

Format CHR$ (aexpr)

CHR$ returns the character with the ASCII code of aexpr.

Example PRINT CHR$ (65)
A

DIM

Execution mode imm &def

Format DIM var subscript [{,Var subscript}]

Upon the execution of a DIM statement, the computer saves space for the array with the name var. In addition, the amount of memory space allocated for an array is indicated by subscript.

Example DIM B% (7)
(The variable B% is declared as an array and the space for a total of 8 integer elements is allocated.)

LEFT$

Executin mode imm & def

Format LEFT$ (sexpr, aexpr)

Left$ returns the leftest /aexpr/ characters of the string sexpr.

Example]PRINT LEFT$ ("MPF",2)
MP

LEN

Execution mode imm & def

Format LEN (sexpr)

LEN returns the length of the string sexpr.

```
Example    A$ = "MICRO PROFESSOR"
           ]PRINT LEN (A$), LEN ("MPF")
            15          3
```

MID$

Execution mode   imm &def

Format   MID$ (sexpr, aexpr 1 [,aexpr 2])

MID$ returns /aexpr 2/ characters starting from the /aexpr 1/-th character of the string sexpr.

```
Example    ] 10 A$ = "MICROPROFESSOR"
           ] 20 PRINT MID$ (A$,4,3)
           ] RUN
             ROP
```

RIGHT$

Execution mode   imm & def

Format   RIGHT$ (sexpr, aexpr)

RIGHT$ returns the rightest /aexpr/ characters of the string sexpr.

```
Example    ]PRINT RIGHT$ ("COMPUTER",4)
           UTER
```

STORE and RECALL

Execution mode   imm &def

Format   STORE avar
          RECALL avar

These two commands store and recall arrays onto/from cassette tape. The dimensions of the array specified by the RECALL statement must be the same as the dimensions of the array designated by the STORE statement.

```
Example    10 DIM A (4,4,4)
           20 A (0,0,0) = 1
           30 A (0,0,1) = 2
           40 A (0,0,2) = 3

           130 STORE A
```

Press PLAY and RECORD buttons on the recorder and key in RUN through the keyboard and you can save the data of array A onto the tape.

If you want to load the data into the MPF-III, type

```
DIM A (4,4,4)
RECALL A
```

VAL

Execution mode imm &def

Format VAL (sexpr)

The VAL command convert a string into a real or integer and returns the value of that number.

Example
```
]10 A = VAL ("-7.4")
]20 PRINT A
]RUN
-7.4
```

3.3.4 Commands and Statements Relating to Input/Output

DATA

Execution mode def

Format : DATA [literal | string | real | integer][{,[literal | string | real | integer]}]

The DATA statement creates and contain a list of elements which are to be retrieved by READ statements. The variable types of corresponding items in DATA and READ statements must match pair by pair.

Example
```
]10 READ A,B,C$
]20 PRINT A ; C$
]30 DATA 3,4, TEK
]RUN
3TEK
```

DEF FN

Execution mode def

Format FN name (aexpr2)

The DEF FN statement allows functions to be defined and used by the user in BASIC programs.

Example

```
]20 DEF FN A(Y) = 8*Y+4
]30 PRINT FN A(8)
]RUN
68
```

GET

Execution mode def

Format GET var

The GET command takes a character from the keyboard without displaying it on the TV screen and does not require the user to press the RETURN key.

Example

```
10 GET A$
20 PRINT A$
] RUN
```

INPUT

Execution mode def

Format INPUT [string;] var[{,var}]

Upon execution of an INPUT statement, MPF-III will wait until it receives input from the keyboard. If the optional string in Format above is not given, then the INPUT statement causes MPF-III to print a question mark and wait for you to key in.

Be sure to press the [↵] key after you key in as required.

Example

```
10 INPUT A,B
20 PRINT A, B, A +B
] RUN
?3, 4
3       4      7
```

IN #

Execution mode imm & def

Format IN# aexpr

The IN# command is used to inform the MPF-III that data will be input through slot /aexpr/. If aexpr=Ø, data will be input through the keyboard. If aexpr = 1 through 7, the various peripheral devices is specified.

Example 1Ø IN # Ø

LET

Execution mode imm & def

Format [LET] avar [subscript] = aexpr
[LET] svar [subsript] = **sexpr**

This statement assigns the value of the expression or string on the right-hand side of the equal sign to the variable name on the lefe.

Example LET A = 3

PRINT

Execution mode imm & def

Format PRINT [{expr}[{,|;[{expr}]}]] [.|;]
PRINT {;}
PRINT {,}

You may use ? in place of PRINT.

The PRINT command is used to output characters to the screen or other outut devices. If you simply type

PRINT

without any options, then a line feed and a return will be outut. However, if PRINT is used with options, the value of the expressions contained in the PRINT statement will be printed. Spacing between printed values are determined by the use of semicolons or commas in a PRINT statement. If semicolons are used, then the next value to be printed is printed immediately following the last value printed. If commas are used, then the next value to be printed is printed in the first position of the next available tab field.

```
Example     10 PRINT "MY PERSONAL COMPUTER"; " is";" MPF"
            ] RUN
            MY PPERSONAL COMPUTER IS MPF
```

PR#

Execution mode   imm & def

Format   PR# aexpr

The PR# command is used to select a output slot as indicated by aexpr, which must be in the range zero through seven. PR# 0 selects the video screen for output.

```
Example     10 PRINT# 0
            20 PRINT "ABC"
           ]RUN
           ABC
```

READ

Execution mode   imm & def

Format   READ var [{, var}]

Upon execution of a program, the value of the first variable in the first READ statement will take on the value of the first element contained in the DATA list, which is comppposed of all the elements in all the DATA statements. If there is a second variable, it will be assigned the value of the second element in the DATA list; the third variable will take on the value of the third element and so on. The READ command takes out items from the DATA list and assigns them to the variables in the READ statement one by one.

```
Example     10 READ A,B,C,
            20 DATA 5, 6, 7
            30 PRINT A * B + C
            ] RUN
            37
```

RESTORE

Execution mode   imm & def

Format   RESTORE

The RESTORE command will move the data list pointer to the beginning of the data list. RESTORE must be used together with READ and DATA in a program.

Example

```
10 READ A, B, C
20 RESTORE
30 READ D,E,F
40 PRINT A; B; C; F; E; D
50 DATA 1, 2, 3
]RUN
123321
```

3.3.5 Commands and Statements Relating to Program Execution Control

FOR .... TO STEP

Execution mode imm & def

Format: FOR real avar = aexpr 1 To aexpr2[STEP aexpr 3]

The FOR....TO...STEP instruction can be used to repeatedly execute a loop, which contains a set of instructions, until /aexpr 2/ reaches a certain value. Upon execution of FOR, avar is given the value of /aexpr1/. The statements following the FOR instruction are then executed until MPF-III encountes a

NEXT avar

where the avar in the NEXT statement is the same as the avar in the FOR statement. /avar/ is then incremented by the value of /aexpr 3/ (or if /aexpr 3/) is not given, avar is automatically incremented by 1). After avar has been incremented, it is then compared to the value of /aexpr 2/ and program execution will proceed as follows :

* If avar is greater than /aexpr2/, then program execution will resume at the statement following NEXT
* If avar is less than or equal to /aexpr2/, the execution will resume at the statement following FOR.

```
Example     10 FOR A = 1 TO 5 STEP 2
            20 PRINT A
            30 NEXT A
            ]  RUN
            1


            3


            5
```

GOSUB

Execution mode   imm & def

Format   GOSUB linenum

This command causes the program to branch to the line given by linenum. If a RETURN statement is encountered, the program will branch back to the line immediately following GOSUB.

```
Example     10 PRINT "MAIN PROGRAM"
            20 GOSUB 100
            30 PRINT "THE END"
            40 END
            100 PRINT "SUBPROGRAM"
            110 RETURN
           ]RUN
           MAIN PROGRAM
           SUBPROGRAM
           THE END
```

GOTO

Execution mode   imm & def

Format   GOTO linenum

The GOTO statement cause the program to branch to the line whose line number is indicated by linenum.

```
Examle     10 PRINT "ABC"
           20 GOTO 10
           30 PRINT "DEF"
          ]RUN
           ABC
           ABC
           ABC
```

IF THEN IF .... GOTO

Execution mode imm & def

```
Format    IF expr THEN instruction [{:instruction}]
          IF expr THEN [GOTO] linenum
          IF expr[THEN] GOTO linenum
```

The IF statement can be used to conditionally test a supposition and make the program execute all of the specified instructions in the THEN portion of the statement if the condition in the IF portion of the statement is true; Otherwise the THEN portion is ignored.

```
Example   10 A=B+3
          20 IF A>50 THEN GOTO 100
          30 PRINT A

             .
          100 A=2A+3B
```

NEXT

Execution mode imm & def

```
Format    NEXT [,avar]
          NEXT [{,avar}]
```

NEXT is used to terminate the FOR...NEXT loop started by a FOR statement. If several avars are used in a NEXT statement, they must be listed in the proper order so that a FOR...NEXT loop is nested within another FOR...NEXT loop.

```
Example   10 FOR A=1 TO 3
          20 PRINT A
          30 NEXT A
         ]RUN
          1
          2
          3
```

ON .... GOTO and ON .... GOSUB

Execution mode def

Format ON aexpr GOTO linenum {[,linenum]}
ON aexpr GOSUB linenum {[,linenum]}.

These two statements both branch to the line number given by the /aexpr/-th item in the list of linenum in the statement after either GOTO or GOSUB.

Example

```
10 A = 1
20 ON A GOTO 50, 60, 70
50 REM A+3

60 REM A-3
70 REM A*3
```

ONERR GOTO

Execution mode def

Format ONERR GOTO linenum

The ONERR GOTO statement causes the program to branch to the given line number indicated by linenum when an error occurs. In order for ONERR GOTO to function properly in your program, it must be executed before an error occurs so that it can be used to avoid printing error message and to avoid stopping program execution.

Example

```
30 ONERR GOTO 70
40 INPUT I
50 PRINT 3/I
60 STOP
70 PRINT "DATA EQUAL TO ZERO"
80 GOTO 40
```

POP

Execution mode imm & def

Format POP

POP has the same effect as a RETURN command except that it will not branch to the statement which immediately follows the last GOSUB statement executed.

```
Example     10 PRINT "MAIL PROGRAM"
            20 GOSUB 50
            30 PRINT "THE END"
            40 END
            50 GOSUB 80
            60 PRINT "SUBPROGRAM 1"
            70 RETURN
            80 PRINT "SUBPROGRAM 2"
            90 POP
           100 RETURN
           ]RUN
            MAIN PROGRAM
            SUBPROGRAM 2
            THE END
```

RESUME

Execution mode   def

Format   RESUME

When RESUME is encountered at the end of an error-handling routine, it will cause the program to continue execution at the beginning of the statement in which the error was encountered.

```
Example     10 ONERR GOTO 60
            20 INPUT A
            30 B = 1/A
            40 PRINT A,B
            50 GOTO 20
            60 PRINT "DATA EQUAL TO ZERO"
            70 INPUT A
            80 RESUME
            ]RUN
            ?0
            DATA EQUAL TO ZERO
            ?2
            2               .5
            ?
```

RETURN

Execution mode   imm & def

Format   RETURN

Execution of RETURN cauese the program to branch to the statement which immediately comes after the last GOSUB statement executed.

Example

```
  10 PRINT "MAIN PROGRAM"
  20 GOSUB 100
  30 PRINT "THE END"
  40 END
 100 PRINT "SUBPROGRAM"
 110 RUN
]RUN
MAIN PROGRAM
SUBPROGRAM
THE END
```

3.3.6 Commands and Statements Relating to Graphics and Game Control

TEXT

Execution mode   imm & def

Format   TEXT

TEXT is used to return to the text mode from the low or high resolution graphics mode.

Example   TEXT

COLOR

Execution mode   imm & def

Format   COLOR   aexpr

COLOR is used to set the color in the low-resolution grahics mode. If the value /aexpr/ is a real number, MPF-III will first convert it to an integer. The value of aexpr must be in the range 0 through 15. (See Fig. 3-3)

Example   COLOR = 2

GR

Execution Mode   imm & def

Format : GR

The GR command converts the screen to the low-resolution graphics mode and leaves four lines for text at the bottom of the screen. When this command is executed, the screen is cleared to black and the cursor is moved to the text window. The execution of GR also sets COLOR to 0.

Example GR

HLIN

Execution mode imm & def

Format HLIN aexpr 1, aexpr 2 AT aexpr 3

The HLIN command can be used to draw a horizontal line from (/aexpr 1/,/aexpr 3/) to (/aexpr 2/, /aexpr 3/) on the screen in the low-resolution graphics mode.

Example
```
10 GR
20 COLOR = 2
30 HLIN 10, 20 AT 30
```

PLOT

Execution mode imm & def

Format PLOT aexpr 1, aexpr 2

The PLOT command is used to plot a dot specified by the location (/aexpr 1/, /aexpr 2/) in the low-resolution graphics mode using the color designated by the last COLOR statement executed.

Example
```
10 GR
20 COLOR = 3
30 PLOT 20,24
```

SCRN

Execution mode imm & def

Format SCRN (aexpr 1, aexpr 2)

SCRN can be used to display the color code of a particular plotted dot.

Example
```
10 GR
20 COLOR = 5
30 PLOT 30,20
40 PRINT SCRN (30, 20)
```

VLIN

Execution mode imm & def

Format VLIN aexpr 1, aexpr 2 AT aexpr 3

The VLIN command is used to draw a vertical line on the screen in the low-resolution graphics mode. **When** executed, VLIN will draw a vertical line from the given x-coordinate /aexpr 1/ to x-coordinate /aexpr 2/ at the y-coordinate /aexpr3/.

Example VLIN 3, 8 AT 2Ø

HCOLOR

Execution mode imm & def

Format HCOLOR = aexpr

The HCOLOR statement is used to set the color for plotting in **high-resolution graphics mode**. The value of /aexpr/ must be in the range Ø through 7. Color names and their corresponding color codes are as follows:

CODE	COLOR	CODE	COLOR
Ø	BLACK 1	4	BLACK 2
1	GREEN	5	ORANGE
2	PURPLE	6	BLUE
3	WHITE 1	7	WHITE 2

Example HCOLOR 2

HGR

Execution mode imm & def

Format HGR

The HGR command is used to change the screen to the high-resolution graphics mode and leaves only four lines visible for text at the bottom of the screen.

Example HGR

HGR2

Execution mode imm & def

Format HGR2

The HGR2 command is used to convert the screen to the full-screen high-resolution graphics mode. HGR2 functions much the same way as HGR.

Example HGR2

HPLOT

Execution mode imm & def

Format HPLOT aexpr 1, aexpr 2 TO aexpr 3, aexpr 4

HPLOT is used in the high-resolution graphics mode to plot horizontal and vertical lines.

Example
```
10 HGR
20 HCOLOR = 1
30 HPLOT 0, 0 TO 279, 159
```

PDL

Execution mode imm & def

Format PDL (aexpr)

PDL returns the current value of the game-controlling paddle (the joystick) specified by aexpr.

Example PDL (0)

DRAW

Execution mode imm & def

Format DRAW aexpr 1, AT aexpr 2, aexpr 3

The DRAW command is used to draw high-resolution graphics shapes starting at the point given by /aexpr 2/ as the x-coordinate and /aexpr 3/ as the y-coordinate. Before the DRAW command can be properly executed, the shape table must have already been loaded using the SHLOAD command.

Example DRAW 3 AT 25, 30

XDRAW

Execution mode imm & def

Format XDRAW aexpr 1 AT aexpr 2, aexpr 3

This **command** can be used to draw a high-resolution graphics shape on the screen and is basically the same as the DRAW command with the following exceptions:

* When XDRAW is first used to draw a shape and then used a second time with the same parameters, it can be used to easily erase the shape without erasing anything is the background;
* When XDRAW is used to draw a shape, it uses the complement of the color used to plot the existing points on the screen. The following pairs are complementary colors:

BLACK and WHITE
BLUE and GREEN

Example **XDRAW** 10 AT 20, 30

ROT

Execution mode imm & def

Format ROT = aexpr

This statement sets the orientation of high-resolution graphic shapes which are drawn by DRAW or XDRAW. The angular rotation used to draw a shape is determined by the value of /aexpr/, which must be a value in the range 0 through 255

Example ROT = 16

SCALE

Execution mode imm & def

Format SCALE = aexpr

SCALE is used to set the size of high-resolution graphic shapes drawn by DRAW or XDRAW.

Example SCALE = 5

SHLOAD

Execution mode   imm & def

Format   SHLOAD

This command loads a high-resolution graphics shape table from cassette tape. When a shape table is to be loaded in memory, it is loaded immediately below HIMEM: and HIMEM: is in turn set immediately bolow the shape table to protect it.

Example   SHLOAD

3.3.7 Built-in Arithmetic Functions

All functions may be used wherever an expression of the same type may be used. They may be used either in immediate or deferred execution mode. Here are brief descriptions of some of arithmetic functions. Other functions are described in sections dealing with similar instructions.

SIN (aexpr)
Returns the sine of \aexpr\ radians.

COS (aexpr)
Returns the cosine of \aexpr\ radians.

TAN (aexpr)
Returns the tangent of \aexpr\ radians.

ATN (aexpr)
Returns the arctangent, in radians, of \aexpr\. The angle returned is in the range -pi/2 through +pi/2 radians.

INT (aexpr)
Returns the largest integer less than or equal to \aexpr\.

RND (aexpr)
Returns a random real number greater than or equal to 0 and less than 1.

If \aexpr\ is greater than zero, RND(aexpr) generates a new random number each time it is used.

If \aexpr\ is less than zero, RND(aexpr) generates the same random number each time it is used with the same \aexpr\, as if from a permanent random number table built into the MPF-III. If a particular negative argument is used to generate a random number, then subsequent random numbers generated with positive arguments will follow the same sequence each time. A different random sequence is initialized by each different negative argument. The primary reason for using a negative argument for RND is to initialize (or "seed") a repeatable sequence of random numbers. This is particularly helpful in debugging programs that use RND.

If \aexpr\ is zero, RND(aexpr) returns the most recent previous random number generated (CLEAR and NEW do not affect this). Sometimes this is easier than assigning the last random number to a variable in order to save it.

SGN (aexpr)
Returns -1 if \aexpr\<Ø, returns Ø if \aexpr\=Ø, and returns 1 if \aexpr\>Ø.

ABS (aexpr)
Returns the absolute value of \aexpr\, that is, \aexpr\ if \aexpr\>=Ø, and -\aexpr\ if \aexpr\ <Ø.

SQR (aexpr)
Returns the positive square root. This is a special implementation that executes more quickly than Ø.5.

EXP (aexpr)
Raises e (to 6 places, e=2.718289) to the indicated power, \aexpr\.

LOG (aexpr)
Returns the natural logarithm of \aexpr\.

3.3.8 Derived Functions

Listed below are some functions which are not provided by the MPF-III BASIC but can be derived from the built-in functions and can be easily implemented by using the DEF FN statement.

SECANT:
SEC(X) = 1/COS (X)

COSECANT:
CSC (X) = 1/SIN(X)

COTANGENT:
COT(X) = 1/TAN(X)

INVERSE SINE:
ARCSIN(X) = ATN(X/SQR(-X*X+1))

INVERSE COSINE:
ARCCOS(X) = -ATN(X/SQR(-X*X+1))+1.5708

INVERSE SECANT:
ARCSEC(X) = ATN(SQR(X*X-1)+(SGN(X)-1)*1.5708

INVERSE COSECANT:
ARCCSC(X) = ATN(1/SQR(X*X-1))+(SGN(X)-1)*1.5708

INVERSE COTANGENT:
ARCCOT(X) = -ATN(X)+1.5708

HYPERBOLIC SINE:
SINH(X) = (EXP(X)-EXP(-X))/2

HYPERBOLIC COSINE
COSH(X) = (EXP(X)+EXP(-X))/2

HYPERBOLIC TANGENT:
TANH(X) = -EXP(-X)/(EXP(X)+EXP(-X))*2+1

HYPERBOLIC SECANT:
SECH(X) = 2/(EXP(X)+EXP(-X))

HYPERBOLIC COSECANT:
CSCH(X) = 2/(EXP(X)-EXP(-X))

HYPERBOLIC COTANGENT:
COTH(X) = EXP(-X)/(EXP(X)-EXP(-X))*2+1

INVERSE HYPERBOLIC SINE:
ARCSINE(X) = LOG(X+SQR(X*X+1))

INVERSE HYPERBOLIC COSINE:
ARCCOSH(X) = LOG(X+SQR(X*X-1))

```
INVERSE HYPERBOLIC TANGENT:
ARCTANH(X) = LOG((1+X)/(1-X))/2

INVERSE HYPERBOLIC SECANT:
ARCSECH(X) = LOG((SQR(-X*X+1)+1)/X

INVERSE HYPERBOLIC COSECANT:
ARCCSCH(X) = LOG(SGN(X)*SQR(X*X+1)+1)/X

INVERSE HYPERBOLIC COTANGENT:
ARCCOTH(X) = LOG((X+1)/(X-1))/2

A MOD B
MOD(A) = INT((A/B-INT(A/B))*B+.05)*SGN(A/B)
```

3.4 Screen Editor

We have given you a short description on some keys for screen editing purposes in 1.3.3. Refer to that section if you have forgot how they work. Below is a more detailed description on the screen editor.

*ESC

The ESC-A(B,C,D) combination is different from ESC-I(J,K,M,) in that ESC-A can move the cursor for only one position and if you want to move further, you have to repeat the ESC-A two-key sequence. While with the ESC-I combinations, you have to press ESC only once (to enter into the movement mode) and then you can move the cursor with I,J,K, and M continually without the need to press ESC every time. In this case, if you press any key other than I,J,K,M, you will exit the movement mode and return to the normal key-in mode.

* [→]

The [→] key moves the cursor to the right while recording the characters you typed onto the memory, if the MPF-III is in movement mode, you must press the space key to exit the movement mode before you expect the --> key to copy characters onto the memory.

* [←]

This is the backspace key used to correct the mistyped characters, it moves the cursor to the left and as it passes, the characters you typed will be deleted from the memory.

* [↓]

The cursor is moved down one position each time this key is pressed.

* [↑]

The [↑] key can move the cursor up only after you have pressed the ESC key once(or odd times after the MPF-III is powered on). A subsequent press on the ESC will disable the [↑] as a screen editing key. In other words, the ESC key functions as a toggle switch as far as the screen editing feature of [↑] is concerned. In 1.3.4 we have also described the two-key sequences of ESC-E(F,@), refer to that section if you have forgot how they work.

To get you familiarized with the screen editor, we give you a few examples below.

Example 1

Key in the following program:

```
]10 PRIMT "TAIPEI OF TAIWAN"
]20 GOTO 10
RUN
 ?SYNTAX ERR IN 10
]◆
```

In the preceding program, we have mistyped the word PRIMT which should be PRINT. In this case, press ESC and then press I to move the cursor up to the character 1 on line 10. Press the space key to exit the movement mode and then press the [→] key to copy the program line until it is on the character M. Key in N and the cursor will move onto T. Press the [→] key again until it is on the position immediately after the second quote in line 10. Finally press [↵]

```
]LIST
10 PRINT "TAIPEI OF TAIWAN"
20 GOTO 10
]◆
```

As shown, you may list the program with the LIST command to make sure the correction is successfully done. Now execute the program.

```
]RUN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN

]CTRL-C

BREAK IN 1Ø
]
```

The program will have "TAIPEI OF TAIWAN" displayed on the screen indefinitely until you press CTRL-C. BREAK IN 1Ø will appear as shown in the example.

Example 2

In example 1, suppose we want to insert TAB(1Ø) after PRINT in line 1Ø. See how we can do this.

First, list the program line we want to edit.

```
]LIST 1Ø
1Ø PRINT "TAIPEI OF TAIWAN"
]
```

Press ESC-D (or I) to move the cursor up and press SPACE and until the cursor is on the first quote. Press ESC-D to move the cursor up above the first quote This is what you will see on the screen:

```
]LIST 1Ø

 1Ø PRINT "TAIPEI OF TAIWAN"
```

Key in TAB(1Ø) which we want to add to the line.

```
]LIST 1Ø
          TAB(1Ø);
 1Ø PRINT "TAIPEI OF TAIWAN"
```

Press ESC-C to move the cursor down,and you will see:

```
]LIST 1Ø

          TAB(1Ø);
 1Ø PRINT "TAIPEI F TAIWAN"
```

* ↑

The ↑ key can move the cursor up only after you have pressed the ESC key once(or odd times after the MPF-III is powered on). A subsequent press on the ESC will disable the ↑ as a screen editing key. In other words, the ESC key functions as a toggle switch as far as the screen editing feature of ↑ is concerned. In 1.3.4 we have also described the two-key sequences of ESC-E(F,@), refer to that section if you have forgot how they work.

To get you familiarized with the screen editor, we give you a few examples below.

Example 1

Key in the following program:

```
]1Ø PRIMT "TAIPEI OF TAIWAN"
]2Ø GOTO 1Ø
RUN
 ?SYNTAX ERR IN 1Ø
]◆
```

In the preceding program, we have mistyped the word PRIMT which should be PRINT. In this case, press ESC and then press I to move the cursor up to the character 1 on line 1Ø. Press the space key to exit the movement mode and then press the → key to copy the program line until it is on the character M. Key in N and the cursor will move onto T. Press the → key again until it is on the position immediately after the second quote in line 1Ø. Finally press ↵

```
]LIST
1Ø PRINT "TAIPEI OF TAIWAN"
2Ø GOTO 1Ø
]◆
```

As shown, you may list the program with the LIST command to make sure the correction is successfully done. Now execute the program.

```
]RUN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN
TAIPEI OF TAIWAN

]CTRL-C

BREAK IN 10
]
```

The program will have "TAIPEI OF TAIWAN" displayed on the screen indefinitely until you press CTRL-C. BREAK IN 10 will appear as shown in the example.

Example 2

In example 1, suppose we want to insert TAB(10) after PRINT in line 10. See how we can do this.

First, list the program line we want to edit.

```
]LIST 10
10 PRINT "TAIPEI OF TAIWAN"
]
```

Press ESC-D (or I) to move the cursor up and press SPACE and until the cursor is on the first quote. Press ESC-D to move the cursor up above the first quote This is what you will see on the screen:

```
]LIST 10

 10 PRINT "TAIPEI OF TAIWAN"
```

Key in TAB(10) which we want to add to the line.

```
]LIST 10
         TAB(10);
 10 PRINT "TAIPEI OF TAIWAN"
```

Press ESC-C to move the cursor down,and you will see:

```
]LIST 10

         TAB(10);
 10 PRINT "TAIPEI F TAIWAN"
```

Press ESC-J to move the cursor onto the first quote in line 1Ø.

```
]LIST 1Ø
          TAB(1Ø);
 1Ø PRINT ⬟TAIPEI OF TAIWAN"
```

Press [→] to move the cursor until it is positioned immediately after the second quote.

```
]LIST 1Ø
          TAB(1Ø);
 1Ø PRINT "TAIPEI OF TAIWAN"⬟
```

Press [↵] and type LIST. Finally press [↵].
Now you will see the final result:

```
]LIST
 1Ø PRINT TAB(1Ø);"TAIPEI OF TAIWAN"
 2Ø GOTO 1Ø
```

Example 3
Again we will use example 1 to see how the two-key sequence ESC-E,F,@ work.

```
]LIST
 1Ø PRINT "TAIPEI OF TAIWAN"
 2Ø GOTO 1Ø
]RUN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN

 BREAK IN 1Ø  (Press CTRL-C to suspend the execution)

]⬟
```

Press ESC-I to move the cursor onto the character R, and then press E. You will see the display on the screen:

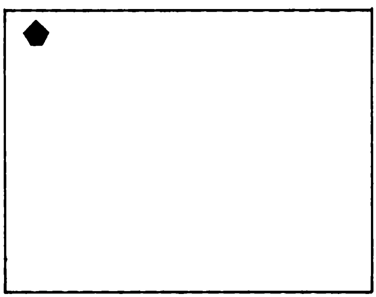
```
 10 PRINT "TAIPEI OF TAIWAN"
 20 GOTO 10
]
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
 TAIPEI OF TAIWAN
```

As is shown, RUN is cleared and the cursor remains at the original position.

Press ESC-F now . You will see the data from the cursor position to the end of the screen is blanked out as shown below.

```
 10 PRINT "TAIPEI OF TAIWAN"
 20 GOTO 10
]
```

Finally press @ .The whole screen is cleared and the cursor is positioned at the upper leftest corner of the screen.



In addition to the fundamental functions stated above, the MPF-III screen editor have the following special features in the 80-column text display mode.

*CHARACTER INSERT(INSC):

INSC can be used to insert a character during program editing.

Example 1:
]10 PRINT "HAPPY YEAR"

In the above line, we have the word NEW missing. See how we can add the word to the line.

Press ESC-I to move the cursor onto the character 1 in the line . Press any key other than I,J,K,M,↑,→,↓,←to exit the movement mode. Use the → key to move the cursor until it is positioned on the character Y of the word YEAR .Key in INSC,N, INSC,E, INSC,W, INSC in that order. Note every time INSC is pressed, the word YEAR is shifted to the right one position. After insertion is finished, press CPES to copy the following data into the edit buffer.

List the edited line with the LIST command:

```
]LIST
 10 PRINT "HAPPY NEW YEAR"
 20 GOTO 10
]
```

*CHARACTER DELETE(DELC):

The DELC key is used to delete a character from the lines.

Example 2

Suppose we want to delete the word HAPPY from line 10 in the preceding example. Repeat the steps in example 1 to move the cursor on the character H this time. Press DELC six times to delete the word HAPPY and the space after the word. Press CPES to copy the following data into the memory. Confirm the result with the LIST command.

```
]LIST
 10 PRINT "NEW YEAR"
 20 GOTO 10
```

*CPES(Copy to end of statement)

CPES will copy the characters from the cursor position to the end of the statement into the edit buffer. CPES, together with INSC and DELC, will save time when you want to recover minor mistakes in the program as shown in these examples.

*CLRS(Clear whole screen):

The CLRS key is used to clear the whole screen and the cursor will remain where it is. It is different from HOME in that with HOME the cursor will be placed at the upper leftest corner and the screen is not cleared.

3.5 SG Music Commands

The MPF-III makes it possible for the user to produce musical pieces by himself. It provides the users with six music commands of BASS, SONG, PLAYINSTR TEMPO and EFFECT.

Make use of these commands together with other BASIC commands, you can fully enjoy composing music

The six music commands are described as follows

(1) SONG

Format   SONG P1, P2

Description   P1: Tone. P2: Duration (unit beat)

Table 3-1 and 3-2 shows the corresponding codes of all the tones and duration.

		DO C	RE D	MI E	FA F	SO G	LA A	SI B
LOW		11	12	13	14	15	16	17
TONE	#SHARP	21	22	*	24	25	26	*

		DO	RE	MI	FA	SO	LA	SI
MIDDLE		31	32	33	34	35	36	37
TONE	#SHARP	41	41	*	44	45	46	*

		DO	RE	MI	FA	SO	LA	SI
HIGH		51	52	53	54	55	56	57
TONE	#SHARP	61	62	*	64	65	66	*

REST	MUSIC	END CODE
	BASIC	ASSEMBLY
ØØ	1ØØ	FF

Table 3-1 The Tones

BEAT	1/16	1/8	¼	½	¾	1	1½	2	3	4
NOTE	𝅘𝅥𝅱	𝅘𝅥𝅰	𝅘𝅥𝅯	♪	♪.	♩	♩.	𝅗𝅥	𝅗𝅥.	𝅝
CODE	70	71	72	73	74	75	76	77	78	79

Table 3-2 The Beats

Format BASS P1

Description P1 selects the kinds of rhythms with a value between zero and 8.The default value is zero. Consult the following table.

Ø	STOP		
1	WALTZ	5	CHACHA
2	RUMBA	6	BLUES
3	TANGO	7	ROCK
4	MARCH	8	SWING

(3) TEMPO

Format Tempo P1

Description

The default value is Tempo 6.

TEMPO	BEAT/MINUTE	TEMPO	BEAT/MINUTE
1	640	9	90
2	320	10	80
3	240	11	70
4	200	12	60
5	160	13	58
6	120	14	55
7	110	15	52
8	100		

(4) INSTR

Format INSTR P1

Description P1 : 1--4
1 = Piano
2 = Bell
3 = Xylophone
4 = Organ
The default value is 1 (piano).

(5) PLAY

Format PLAY P1, P2

Description P1 The memory location of the table.
P2 Select BASS

PLAY differs from SONG in that PLAY assigns a section of memory locations as the buffer while SONG fetches data directly from DATA statement.

(6) EFFECT

Format EFFECT P1

Description P1 0--3
0 dropping a bomb
1 explosion
2 raser gun
3 machine gun

Below we present a list of programs to show you how to use these music commands.

Example 1 SONG 31, 75
(DO one beat)

```
Example 2     10 SONG 31, 75
              20 SONG 32, 75
              30 SONG 33, 75
              40 SONG 34, 75
```

A sequence of four notes DO, RE, MI, FA are played.

```
Example 3     10 FOR H=1 TO 14
              20 READ A, B
              30 SONG A, B
              40 NEXT
              50 DATA   31,75,32,75,33,75,34,75,35,75,36,
                        75,37,75,51,75,52,75,53,75,54,75,
                        55,75,56,75,57,75
```

A total of 14 notes are played.

Try the beautiful melody produced by the following program.

```
Example 4     5 BASS   2
              10 FOR   H=1 TO 40
              20 READ  A, B
              30 SONG  A,B
              40 NEXT
              50 DATA  33,75,36,73,36,75,33,75,32,75,31,
                       73,32,73,31,73,17,73,16,75,13,75,
                       16,75,17,75,31,75,17,75,16,73,17,
                       73,31,73,32,73,33,78.33,75,36,73,
                       36,73,33,75,32,75,31,73,32,73,31,
                       73,17,73,16,75,33,75,16,75,17,75,
                       31,75,17,75,16,75,16,73,16,73,16,
                       78,100,100,
```

Try the following sequence of special sound effects.

```
Example 5    10   EFFECT  0
             20   EFFECT  1
             30   EFFECT  2
             40   EFFECT  1
             50   EFFECT  3
             60   EFFECT  1
```

In the last example first key in the music score of the popular song in memory starting from $5000 (hexadecimal)

Example 6

```
]CALL-151 [↵]

*5000     5000    31 73 32 73 33 73 31 73 31
                  73 32 73 33 73 31 73
*5010     5010    33 73 34 73 35 73 33 73 34
                  73 35 73 35 72 36 72
*5020     5020    35 72 34 72 33 73 31 73 35
                  72 36 72 35 72 34 72
*5030     5030    33 75 31 75 31 75 15 75 31
                  75 31 75 15 75 31 75
*5040     5040    FF FF

* [CTRL] [C]
```

Now see how the MPF-III will play it;

```
]10   A = 5*4096 + 0*256 + 0*16 +0
]20   PLAY  A,2
]RUN
```

Or you can try the song playing in xylophone with a BLUES rhythm.

```
]10   TEMPO  3
]20   INSTR  3
]30   PLAY 20480,6
]RUN
```

Here 20480 is the decimal of 5000H.

Chapter 4

MORE ON THE OPTIONAL DEVICES

4.1 Games on the Cassette Tape

The MPF-III provides a wide variety of games for entertainment which are recorded on magnetic tapes. To load a game program into your MPF-III, you must follow the procedures described below :

(1) Connect one end of the cable to the EAR or PHONE jacket on the cassette recorder and the other end to the EAR jacket on the MPF-III. Adjust the volume so that it is somewhere near the maximum.

(2) Place the cassette into your recorder and rewind the tape to the beginning.

(3) Type LOAD at the keyboard, start the recorder playing and press [↵]'.

(4) After loading is finished, you can start enjoying the games according to the directions on the tapes.

4.2 Serial Interface Card(SIC)

SIC is a programmable circuit board based on a R6551 ACIA (Rockwell's Asynchronous Communication Interface Adapter) chip which is connected to the MPF-III through a cartridge connector for Industrial Standard Asynchronous Serial Data Transmission between the MPF-III and the external devices.

With SIC, the MPF-III can be connected to any device with serial interface (e.g, CRT).

The Procedure for installing SIC is listed below:

(1) Turn off the power and insert the serial interface cartridge. Be sure the pins are in the right place.

(2) SIC is connected to the external devices through the specification of the baud rate. In other words you have to set the DIP switvh to a corresponding baud rate required for the MPF-III to go well with external devices.

(3) Connect SIC and the external device with a cable and the procedure is finished.

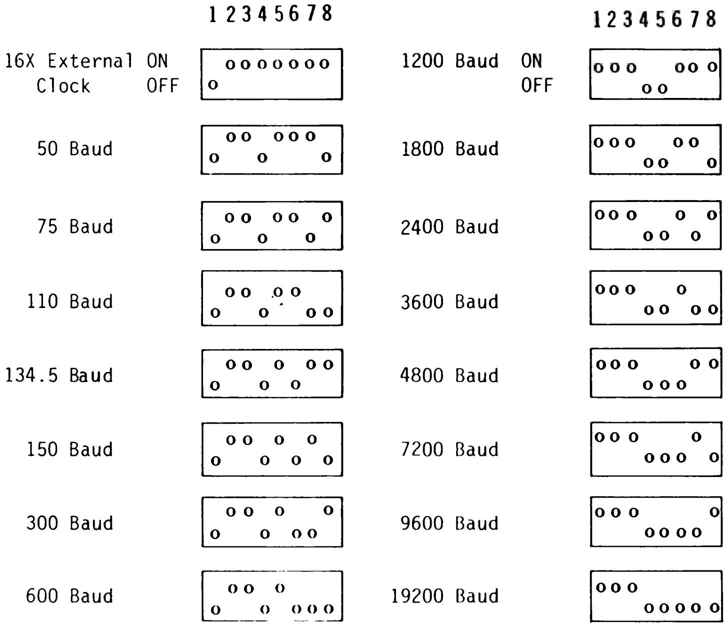


Table. 4-1 Baud rate Select Switch

4.3 Floppy Disk Interface (FDI)

Insert the disk drive interface card into the specified slot on the MPF-III main PC board and plug the drive cable into the connector marked "DRIVE 1' on the back paned. Be sure yu turn off the power before you follow the steps stated above.

After connection is completed, turn on the power.

First you will see "MPF-III" displayed on the screen, and then you will hear the disk drive clicking and whirring, signaling it is reading the data and a few moments afterwards, you will see the prompt appear on the lower left-hand corner on the screen and the disk drive is ready to follow your instructions.

4.4 Printer

Installing the Printer

1. Connect the printer cable and turn on the power.

2. Type PR#1 to start the printer.

3. Type PR#Ø to stop the printer.

Function and Operation of the Printer Driver.

(The following operations are supposed to be executed on the EPSON printer.)

1.Changing the type

* Extend Mode

```
170  PRINT  CHR$ (14);"EXTEND MODE...."
180  PRINT
```

```
EXTEND MODE...
```

Curtail mode

```
190  PRINT  CHR$ (15);"CURTAIL MODE..."
```

```
CURTAIL MODE...
```

Back to normal

```
200  PRINT  CHR$ (18)
```

```
PRINTER DRIVER
```

2. Set linefeed space   (Default is 1/6 inch/line)

ESC, "A", CHR$(I)   where ESC = CHR$(27), I is a variable, linefeed space = I/72

```
205  REM  LINE SPACE CHANGE
210  FOR I = 4 TO 12
220  PRINT  CHR$ (27);"A"; CHR$ (I)
230  PRINT "VARIABLE LINE SPACE"
240  NEXT I
```

```
VARIABLE LINE SPACE
VARIABLE LINE SPACE
VARIABLE LINE SPACE

VARIABLE LINE SPACE

VARIABLE LINE SPACE

VARIABLE LINE SPACE

VARIABLE LINE SPACE

VARIABLE LINE SPACE

VARIABLE LINE SPACE
```

3. Set line length   (Default is 80 characters/line)

CHR$(9); CHR$(48)   10 characters/line   CHR$(9)=CTRL-I
CHR$(9); CHR$(49)   20 characters/line   30 30 CPL
        :   80 40
        :
        :
        :
CHR$(9); CHR$(63)   160 characters/line

"9"   (60) 130 cpl. -> (9) + "/"

```
299  REM  CHANGE LINE LENGTH
300  PRINT "CHANGE LINE LENGTH 80 TO 30"
310 B$ = "XXXXXXXXXOXXXXXXXXXOXXXXXXXXXO
XXXXXXXXXOXXXXXXXXXOXXXXXXXXXO"
320  PRINT B$
330  PRINT  CHR$ (9); CHR$
340  PRINT
350  PRINT B$
360  PRINT  CHR$ (9); CHR$
```

```
CHANGE LINE LENGTH 80 TO 30
XXXXXXXXXOXXXXXXXXXOXXXXXXXXXOXX
XXXXXXXXXOXXXXXXXXXO
```

4. Bit Image:

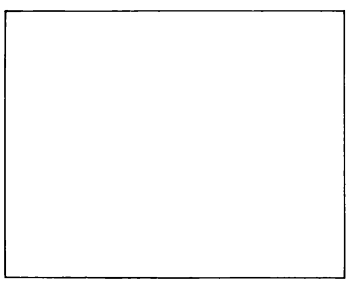
```
75  REM  BIT IMAGE
80  PRINT "BIT IMAGE  ";
85  PRINT  CHR$ (27); CHR$ (75); CHR$ (8); CHR$ (0);
90  PRINT  CHR$ (34); CHR$ (82); CHR$ (138); CHR$ (143);
    CHR$ (138); CHR$ (82) CHR$ (34); CHR$ (0)
```

```
    BIT IMAGE
```


5. HARD COPY :
    PRINT CHR$(17)   CHR$(17)=CTRL-Q

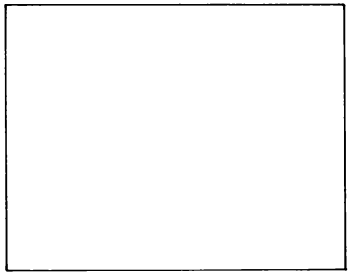
```
]900 HGR: HCOLOR=3
]910 HPLOT 0,0  TO 100,0 TO 100,100 TO 0,100 TO 0,0
]920 HGR2
]930 HPLOT 10,10 TO 8Ø,10 TO 80,80 TO 10,80 TO 10,10
]950 TEXT
]1000 REM GRAPHIC PRINTING
]1015 REM HARD COPY
]1030 PRINT "HARD COPY PAGE 1"
]1050 PRINT CHR$(17)
```

```
        HARD COPY PAGE 1
```



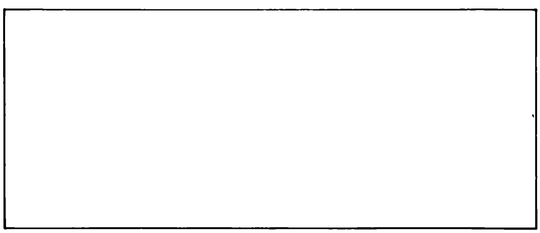
```
1060  PRINT  CHR$ (9);"S"
1070  PRINT "HARD COPY PAGE 2"
1080  PRINT  CHR$ (17)
```

HARD COPY PAGE 2



```
1090  PRINT  CHR$ (9);"P" <---
1135  REM  DOUBLE WIDTH
1150  PRINT "DOUBLE WIDTH PAGE 1"
1155  PRINT  CHR$ (9);"X"
1160  PRINT  CHR$ (17)
11    PRINT  CHR$ (27);"A"; CHR$ (12)·
1180  PRINT  CHR$ (9);"Q"
```

DOUBLE WIDTH PAGE 1



4.5 Joystick

The joystick (see Fig 4-1) can be connected to the MPF-III to facilitate operation in a game. Below is a list of notes on the connection and usage of the joystick:

(1) Insert the joystick cable into the connector marked "JOYSTICK" on the right side of the system unit as shown in Fig 4-2.

(2) The stick is used to control the up, down, left and right movement of the object.

(3) PBØ: Push Button Ø.

(4) PB1: Push Button 1.

(5) U/D (Up/Down) variable resistor:

Used to adjust the range of the vertical positions of the joystick which the system unit can read.

(6) L/R (Left/Right) variable resistor:

Used to adjust the range of the horizontal positions of the joystick the system unit can read.

Note: It is recommended to turn the U/D and L/R to the maximum in a game.

Don't try to separate the stick which was sealed before delivered to you, otherwise you may break it when you use it in a game.

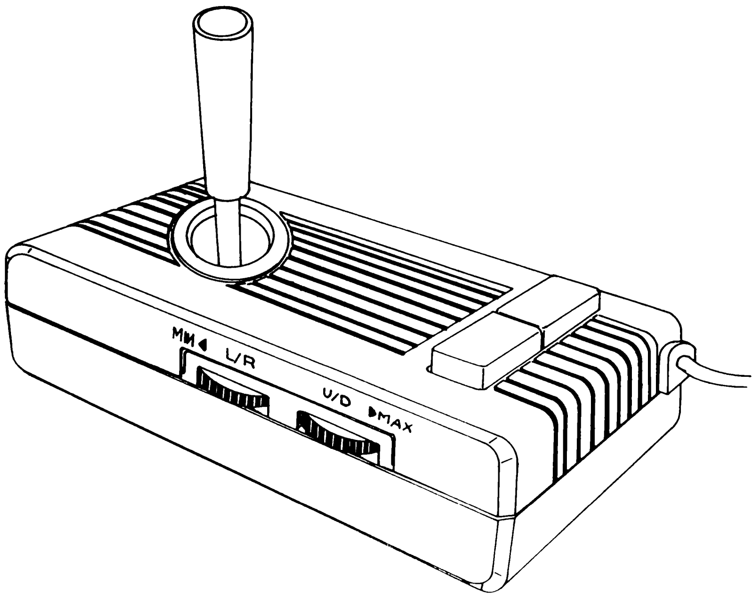


Fig 4-1

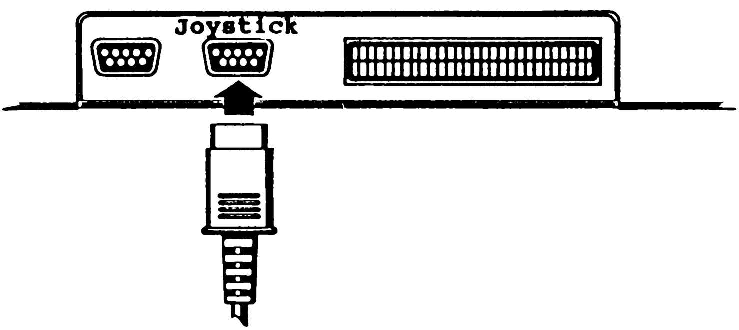


Fig 4-2

Chapter 5

HANDLING THE MPF-III WITH CARE

5.1 General Remarks

Before You Power on the MPF-III

(1) It is always a good practice to check if the system is installed according to the directions before the power is turned on.

(2) Be sure all the cables are connected correctly.

(3) Check if all the switches on the printer and all the accessories are set correctly.

During operation

(1) It is recommended to make backups for all important diskettes.

(2) During program editing, make a habit of saving the file under processing at intervals. This is to prevent unexpected loss of data and programs as a result of power failure or mechanic disorder of the system.

(3) Label every diskette to provide information about the contents for quick reference.

(4) Never open the drive door when the disk drive is in operation.

Before You Turn Off the MPF-III

(1) Make sure you have all the backups for newly made files.

(2) Take the diskettes out of the disk drive before you turn off the power.

(3) Keep a detailed record of the operating conditions to facilitate maintenance of the system.

5.2 Handling the Diskettes with Care

Diskettes are delicate, vulnerable items that require exetreme care in operation, below is a few pieces of advice.

(1) Never place the diskettes near magnetic material such as transformer, screwdriver and other magnetized metal.

(2) If a diskette is not to be used for some time, you better put it in the jacket to protect the diskette from dust, ashes or greasy stains. Never rub, touch, strike or clean the diskette.

(3) Label the diskettes with useful information such as date and contents properly.

(4) Place the original and the copy of a diskette in different places to guard off danger of losing them at the same time.

(5) Never bend or fold the diskettes or expose them to extreme heat in order to protect them from deformation.

(6) Provide the diskettes with a clean and neat environment to diminish danger of damage from dust and stains.

5.3 Dealing with the Files

How to Deal with a Oversized File

When the data of a file is too much to be accommodated by a single diskette, you have to store the file in more than two diskette. When you allocate the parts to different diskettes, do it according to their contents so that you will not use the file with difficulty because of the division.

Damaged Files

Files may be damaged due to the incorrect operation of the user, such as keying in the prohibited control characters. The machine troubles may damage the files as well. As a result, these files can not be loaded or executed. In some cases, you can recover a file with the help of the editing program provided you are familiar with the structure of the file and the functions of the editing program.

There are times when the system diskette is stained or injured and the contents are affected. Some programs may still run normally with such system diskettes. It is, however, quite possible that some of commands can no more be executed and will consequently damage the files. If you find a system diskette with flaws, be sure to replace it right away to ensure a perfect performance of the system.

Copying the Diskettes

Most disk operating systems provide programs for copying diskette. Make it a habit to get backup diskettes for important diskettes.

Deleting Data from Diskette

On the following two occasions you may find it necesary to delete data from a diskette :

(1) The diskette is in good condition and you find some data on it no more useful and want to delete them.

(2) The directory of the diskette is damaged and the data on it can not be retrieved and used.

You may delete data from a diskette as described below:

(1) If the directory of the diskette is in good condition, just use the DELETE command.

(2) If the directory is damaged, you have to use the INIT command to initialize and get a blank, usable diskette.

5.4 Using the Printer with Care

A printer of good quality can be expected to work for years. Abuses of the printer, however, will gravely damage the facility and result in machine troubles.

Listed below is a few notes you are recommended to read through on how to use the printer.

(1) Never use papers of poor quality for printing. They may break or cause obstruction.

(2) Never use ribbons of poor quality. They will damage the printer head. Replace old ribbons when the color is faint.

(3) Clean the printer at times to remove the paper scraps and dusts from within the printer.

(4) Avoid having the printer run for an improperly long time. Stop the machine for a rest at intervals. When you have a lot of material to print.

(5) When you power on the system and find the printer out of order, first check the switch setting at the various location. Make sure the correct printer driver is selected.

5.5 An Ideal Computer Room

The computer is a powerful tool. It is, however, vulnerable to the slightest changes of the external environment. Make a computer room to accommadate your MPF-III according to the suggestions listed below.

(1) Maintain a good ventilation system and keep a stable temperature best suited for the computer.

(2) Provide enough file cabinets for storing diskettes.

(3) Keep a detailed record of the operating conditions of the computer system in the computer room.

(4) Magnetic material and liquid chemicals are prohibited to be stored in the computer room.

(5) Do not carpet the computer room, otherwise the system might be affected by static electricity and dust.

(6) Smoking is prohibited

(7) Never pile heavy installations upon the system.

(8) Bills of notes concerning the operation, maintenance and repair of the computer system are suggested to be pasted on the wall to facilitate the user.

5.6 What's Wrong with Your MPF-III?

Check Out the Mechanic

1. Is the power connected and turned on ? (observe the indicator lamps)

2. Are all the parts located at the right places ?

3. Are all the cables and plugs connected properly ?

Check Out the Commands Typed

1. Power off the MPF-III and power on again.

2. Retype the command.

Check Out the Diskettes

Be sure to make backups for all important diskettes.

Check Out the Software

1.Is the softwre of the right version selected

Check Out the Hardware

(The following steps are recommended to be taken only in the presence of authorized technicians)

1. Turn off the power, remove the cover and take out the PC board. If necessary, remove the connections. Carefully recover the connections after inspection.

2. If the cause of the trouble is verified to be on a certain PC board, remove the components, inspect the joints and restore

3. Replace all possible damaged devices with new ones.

5.7 Diagnostic First-aids for Machine Troubles

Symptom: Power is on but the MPF-III does not work at all.

Diagnosis: Inspect all the connections of the system unit:
1. The power cord.
2. The signal cables.
3. The switches.

Symptom The printer does not work

Diagnosis
1. Check the fuse.
2. See if the Self-test operates properly.
3. Inspect all the switch setting.
4. Check the signal cable for cases of poor connection or breakpoints.
5. Refeed the paper.
6. Keyin the command to start the printer.

Symptom The printer won't stop printing.

Diagnosis
1. Keyin the command to stop printing.
2. Turn off the power.

Symptomn The system does not work

Diagnosis
1. Retype CONTROL C.
2. Power off and power on again.

Symptom Something is wrong with the disk drive.

Diagnosis
1. Make sure the signal cable is firmly connected in the right place.
2. Change the diskette.

Note: The list above is for reference only, for problems you find difficult to resolve, please contact your local MPF-III dealer.

APPENDICES

A 6502 Instructions

ADC	Add Memory to Accumulator with Carry
AND	"AND" Memory with Accumulator
ASL	Shift Left One Bit (Memory or Accumulator)
BCC	Branch on Carry Clear
BCS	Branch on Carry Set
BEQ	Branch on Result Zero
BIT	Test Bits in Memory with Accumulator
BMI	Branch on Result Minus
BNE	Branch on Result not Zero
BPL	Branch on Result Plus
BRK	Force Break
BVC	Branch on Overflow Clear
BVS	Branch on Overflow Set
CLC	Clear Carry Flag
CLD	Clear Decimal Mode
CLI	Clear Interrupt Disable Bit
CLV	Clear Overflow Flag
CMP	Compare Memory and Accumulator
CPX	Compare Memory and Index X
CPY	Compare Memory and Index Y
DEC	Decrement Memory by One
DEX	Decrement Index X by One
DEY	Decrement Index Y by One
EOR	"Exclusive-Or" Memory with Accumulator
INC	Increment Memory by One
INX	Increment Index X by One
INY	Increment Index Y by One
JMP	Jump to New Location
JSR	Jump to New Location Saving Return Address
LDA	Load Accumulator with Memory
LDX	Load Index X with Memory
LDY	Load Index Y with Memory
LSR	Shift Right one Bit (Memory or Accumulator)
NOP	No Operation
ORA	"OR" Memory with Accumulator
PHA	Push Accumulator on Stack
PHP	Push Processor Status on Stack
PLA	Pull Accumulator from Stack
PLP	Pull Processor Status from Stack
ROL	Rotate One Bit Left (Memory or Accumulator)
ROR	Rotate One Bit Right (Memory or Accumulator)
RTI	Return from Interrupt
RTS	Return from Subroutine
SBC	Subtract Memory from Accumulator with Borrow
SEC	Set Carry Flag
SED	Set Decimal Mode
SEI	Set Interrupt Disable Status.
STA	Store Accumulator in Memory
STX	Store Index X in Memory
STY	Store Index Y in Memory
TAX	Transfer Accumulator to Index X
TAY	Transfer Accumulator to Index Y
TSX	Transfer Stack Pointer to Index X
TXA	Transfer Index X to Accumulator
TXS	Transfer Index X to Stack Pointer
TYA	Transfer Index Y to Accumulator

THE FOLLOWING NOTATION APPLIES TO THIS SUMMARY:

A	Accumulator
X, Y	Index Registers
M	Memory
$\bar{C}$	Borrow
P	Processor Status Register
S	Stack Pointer
√	Change
–	No Change
+	Add
∧	Logical AND
—	Subtract
⊻	Logical Exclusive Or
↑	Transfer From Stack
↓	Transfer To Stack
→	Transfer To
←	Transfer To
V	Logical OR
PC	Program Counter
PCH	Program Counter High
PCL	Program Counter Low
OPER	Operand
#	Immediate Addressing Mode

FIGURE 1 ASL-SHIFT LEFT ONE BIT OPERATION

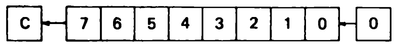


FIGURE 2 ROTATE ONE BIT LEFT (MEMORY OR ACCUMULATOR)

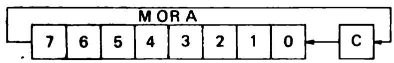
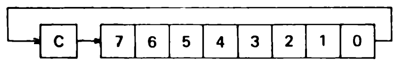


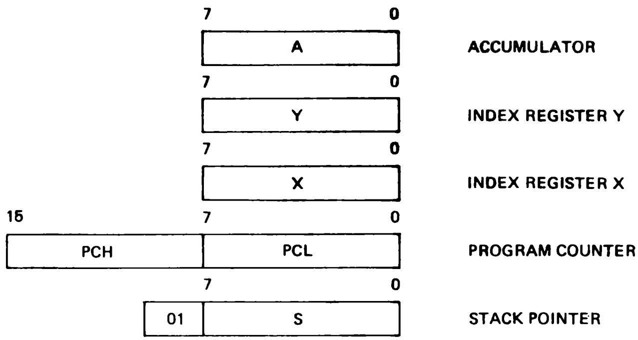
FIGURE 3

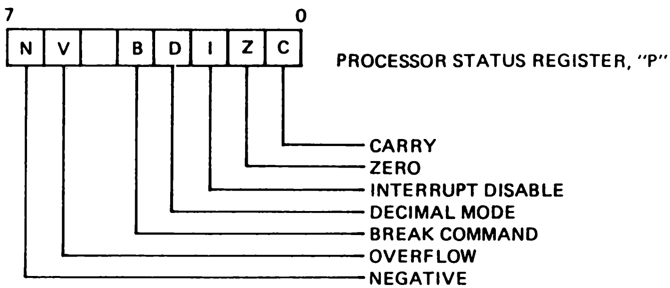


NOTE 1 BIT – TEST BITS

Bit 6 and 7 are transferred to the status register If the result of A ∧ M is zero then Z = 1, otherwise Z = 0

PROGRAMMING MODEL





INSTRUCTION CODES

Name Description	Operation	Addressing Mode	Assembly Language Form	HEX OP Code	No. Bytes	"P" Status Reg. N Z C I D V
ADC Add memory to accumulator with carry	A+M+C → A,C	Immediate	ADC #Oper	69	2	√√√——√
		Zero Page	ADC Oper	65	2	
		Zero Page, X	ADC Oper, X	75	2	
		Absolute	ADC Oper	6D	3	
		Absolute, X	ADC Oper, X	7D	3	
		Absolute, Y	ADC Oper, Y	79	3	
		(Indirect, X)	ADC (Oper, X)	61	2	
		(Indirect), Y	ADC (Oper), Y	71	2	
AND "AND" memory with accumulator	A ∧ M → A	Immediate	AND #Oper	29	2	√√————
		Zero Page	AND Oper	25	2	
		Zero Page, X	AND Oper, X	35	2	
		Absolute	AND Oper	2D	3	
		Absolute, X	AND Oper, X	3D	3	
		Absolute, Y	AND Oper, Y	39	3	
		(Indirect, X)	AND (Oper, X)	21	2	
		(Indirect), Y	AND (Oper), Y	31	2	
ASL Shift left one bit (Memory or Accumulator)	(See Figure 1)	Accumulator	ASL A	0A	1	√√√———
		Zero Page	ASL Oper	06	2	
		Zero Page, X	ASL Oper, X	16	2	
		Absolute	ASL Oper	0E	3	
		Absolute, X	ASL Oper, X	1E	3	
BCC Branch on carry clear	Branch on C=0	Relative	BCC Oper	90	2	——————
BCS Branch on carry set	Branch on C=1	Relative	BCS Oper	B0	2	——————
BEQ Branch on result zero	Branch on Z=1	Relative	BEQ Oper	F0	2	——————
BIT Test bits in memory with accumulator	A ∧ M. M_7 → N. M_6 → V	Zero Page	BIT* Oper	24	2	M_7√———M_6
		Absolute	BIT* Oper	2C	3	
BMI Branch on result minus	Branch on N=1	Relative	BMI Oper	30	2	——————
BNE Branch on result not zero	Branch on Z=0	Relative	BNE Oper	D0	2	——————
BPL Branch on result plus	Branch on N=0	Relative	BPL Oper	10	2	——————
BRK Force Break	Forced Interrupt PC+2 ↓ P ↓	Implied	BRK*	00	1	———1——
BVC Branch on overflow clear	Branch on V=0	Relative	BVC Oper	50	2	——————

Note1: Bit 6 and 7 are transferred tothe status register if the result of AVM is zero then Z=1 otherwise Z=0
Note2: A BRK command cannot be masked by setting

Name Description	Operation	Addressing Mode	Assembly Language Form	HEX OP Code	No. Bytes	"P" Status Reg. N Z C I D V
BVS Branch on overflow set	Branch on V=1	Relative	BVS Oper	70	2	------
CLC Clear carry flag	0 → C	Implied	CLC	18	1	---0--
CLD Clear decimal mode	0 → D	Implied	CLD	D8	1	-0----
CLI	0 → 1	Implied	CLI	58	1	---0--
CLV Clear overflow flag	0 → V	Implied	CLV	B8	1	0-----

CMP
Compare memory and accumulator
	A − M	Immediate
		Zero Page
		Zero Page, X
		Absolute
		Absolute, X
		Absolute, Y
		(Indirect, X)
		(Indirect), Y
			CMP #Oper
			CMP Oper
			CMP Oper, X
			CMP Oper
			CMP Oper, X
			CMP Oper, Y
			CMP (Oper, X)
			CMP (Oper), Y
				C9
				C5
				D5
				CD
				DD
				D9
				C1
				D1
					2
					2
					2
					3
					3
					3
					2
					2
						√√√---

CPX Compare memory and index X	X − M	Immediate Zero Page Absolute	CPX #Oper CPX Oper CPX Oper	E0 E4 EC	2 2 3	√√√---
CPY Compare memory and index Y	Y − M	Immediate Zero Page Absolute	CPY #Oper CPY Oper CPY Oper	C0 C4 CC	2 2 3	√√√---
DEC Decrement memory by one	M − 1 → M	Zero Page Zero Page, X Absolute Absolute, X	DEC Oper DEC Oper, X DEC Oper DEC Oper, X	C6 D6 CE DE	2 2 3 3	√√----
DEX Decrement index X by one	X − 1 → X	Implied	DEX	CA	1	√√----
DEY Decrement index Y by one	Y − 1 → Y	Implied	DEY	88	1	√√----

Name Description	Operation	Addressing Mode	Assembly Language Form	HEX OP Code	No. Bytes	"P" Status Reg. N Z C I D V
EOR "Exclusive-Or" memory with accumulator	$A \veebar M \rightarrow A$	Immediate	EOR #Oper	49	2	√√————
		Zero Page	EOR Oper	45	2	
		Zero Page, X	EOR Oper, X	55	2	
		Absolute	EOR Oper	4D	3	
		Absolute, X	EOR Oper, X	5D	3	
		Absolute, Y	EOR Oper, Y	59	3	
		(Indirect, X)	EOR (Oper, X)	41	2	
		(Indirect), Y	EOR (Oper), Y	51	2	
INC Increment memory by one	$M + 1 \rightarrow M$	Zero Page	INC Oper	E6	2	√√————
		Zero Page, X	INC Oper, X	F6	2	
		Absolute	INC Oper	EE	3	
		Absolute, X	INC Oper, X	FE	3	
INX Increment index X by one	$X + 1 \rightarrow X$	Implied	INX	E8	1	√√————
INY Increment index Y by one	$Y + 1 \rightarrow Y$	Implied	INY	C8	1	√√————
JMP Jump to new location	$(PC+1) \rightarrow PCL$ $(PC+2) \rightarrow PCH$	Absolute	JMP Oper	4C	3	——————
		Indirect	JMP (Oper)	6C	3	
JSR Jump to new location saving return address	$PC+2\downarrow$ $(PC+1) \rightarrow PCL$ $(PC+2) \rightarrow PCH$	Absolute	JSR Oper	20	3	——————
LDA Load accumulator with memory	$M \rightarrow A$	Immediate	LDA #Oper	A9	2	√√————
		Zero Page	LDA Oper	A5	2	
		Zero Page, X	LDA Oper, X	B5	2	
		Absolute	LDA Oper	AD	3	
		Absolute, X	LDA Oper, X	BD	3	
		Absolute, Y	LDA Oper, Y	B9	3	
		(Indirect, X)	LDA (Oper, X)	A1	2	
		(Indirect), Y	LDA (Oper), Y	B1	2	
LDX Load index X with memory	$M \rightarrow X$	Immediate	LDX #Oper	A2	2	√√————
		Zero Page	LDX Oper	A6	2	
		Zero Page, Y	LDX Oper, Y	B6	2	
		Absolute	LDX Oper	AE	3	
		Absolute, Y	LDX Oper, Y	BE	3	
LDY Load index Y with memory	$M \rightarrow Y$	Immediate	LDY #Oper	A0	2	√√————
		Zero Page	LDY Oper	A4	2	
		Zero Page, X	LDY Oper, X	B4	2	
		Absolute	LDY Oper	AC	3	
		Absolute, X	LDY Oper, X	BC	3	

Name Description	Operation	Addressing Mode	Assembly Language Form	HEX OP Code	No. Bytes	"P" Status Reg. N Z C I D V
LSR Shift right one bit (memory or accu- mulator)	(See Figure 1)	Accumulator Zero Page Zero Page, X Absolute Absolute, X	LSR A LSR Oper LSR Oper, X LSR Oper LSR Oper, X	4A 46 56 4E 5E	1 2 2 3 3	0√√———
NOP No operation	No Operation	Implied	NOP	EA	1	——————
ORA "OR" memory with accumulator	A V M → A	Immediate Zero Page Zero Page, X Absolute Absolute, X Absolute, Y (Indirect, X) (Indirect), Y	ORA #Oper ORA Oper ORA Oper, X ORA Oper ORA Oper, X ORA Oper, Y ORA (Oper, X) ORA (Oper), Y	09 05 15 0D 1D 19 01 11	2 2 2 3 3 3 2 2	√√————
PHA Push accumulator on stack	A ↓	Implied	PHA	48	1	
PHP Push processor status on stack	P ↓	Implied	PHP	08	1	
PLA Pull accumulator from stack	A ↑	Implied	PLA	68	1	√√————
PLP Pull processor status from stack	P ↑	Implied	PLP	28	1	From Stack
ROL Rotate one bit left (memory or accu- mulator)	(See Figure 2)	Accumulator Zero Page Zero Page, X Absolute Absolute, X	ROL A ROL Oper ROL Oper, X ROL Oper ROL Oper, X	2A 26 36 2E 3E	1 2 2 3 3	√√√———
ROR Rotate one bit right (memory or accu- mulator)	(See Figure 3)	Accumulator Zero Page Zero Page, X Absolute Absolute, X	ROR A ROR Oper ROR Oper, X ROR Oper ROR Oper, X	6A 66 76 6E 7E	1 2 2 3 3	√√√———

Name Description	Operation	Addressing Mode	Assembly Language Form	HEX OP Code	No. Bytes	"P" Status Reg. N Z C I D V
RTI Return from interrupt	P ↑ PC ↑	Implied	RTI	40	1	From Stack
RTS Return from subroutine	PC ↑, PC+1 → PC	Implied	RTS	60	1	——————
SBC Subtract memory from accumulator with borrow	A - M - $\overline{C}$ → A	Immediate	SBC #Oper	E9	2	√√√——
		Zero Page	SBC Oper	E5	2	
		Zero Page, X	SBC Oper, X	F5	2	
		Absolute	SBC Oper	ED	3	
		Absolute, X	SBC Oper, X	FD	3	
		Absolute, Y	SBC Oper, Y	F9	3	
		(Indirect, X)	SBC (Oper, X)	E1	2	
		(Indirect), Y	SBC (Oper), Y	F1	2	
SEC Set carry flag	1 → C	Implied	SEC	38	1	——1———
SED Set decimal mode	1 → D	Implied	SED	F8	1	————1—
SEI Set interrupt disable status	1 → I	Implied	SEI	78	1	———1——
STA Store accumulator in memory	A → M	Zero Page	STA Oper	85	2	——————
		Zero Page, X	STA Oper, X	95	2	
		Absolute	STA Oper	8D	3	
		Absolute, X	STA Oper, X	9D	3	
		Absolute, Y	STA Oper, Y	99	3	
		(Indirect, X)	STA (Oper, X)	81	2	
		(Indirect), Y	STA (Oper), Y	91	2	
STX Store index X in memory	X → M	Zero Page	STX Oper	86	2	——————
		Zero Page, Y	STX Oper, Y	96	2	
		Absolute	STX Oper	8E	3	
STY Store index Y in memory	Y → M	Zero Page	STY Oper	84	2	——————
		Zero Page, X	STY Oper, X	94	2	
		Absolute	STY Oper	8C	3	
TAX Transfer accumulator to index X	A → X	Implied	TAX	AA	1	√√————
TAY Transfer accumulator to index Y	A → Y	Implied	TAY	A8	1	√√————
TSX Transfer stack pointer to index X	S → X	Implied	TSX	BA	1	√√————

Name Description	Operation	Addressing Mode	Assembly Language Form	HEX OP Code	No. Bytes	"P" Status Reg. N Z C I D V
TXA Transfer index X to accumulator	X → A	Implied	TXA	8A	1	√√----
TXS Transfer index X to stack pointer	X → S	Implied	TXS	9A	1	
TYA Transfer index Y to accumulator	Y → A	Implied	TYA	98	1	√√----

HEX OPERATION CODES

00 – BRK
01 – ORA– (Indirect, X)
02 – NOP
03 – NOP
04 – NOP
05 – ORA– Zero Page
06 – ASL – Zero Page
07 – NOP
08 – PHP
09 – ORA– Immediate
0A – ASL – Accumulator
0B – NOP
0C – NOP
0D – ORA– Absolute
0È – ASL – Absolute
0F – NOP
10 – BPL
11 – ORA– (Indirect), Y
12 – NOP
13 – NOP
14 – NOP
15 – ORA– Zero Page, X
16 – ASL – Zero Page, X
17 – NOP
18 – CLC
19 – ORA– Absolute, Y
1A – NOP
1B – NOP
1C – NOP
1D – ORA– Absolute, X
1E – ASL – Absolute, X
1F – NOP
20 – JSR
21 – AND– (Indirect, X)
22 – NOP
23 – NOP
24 – BIT – Zero Page
25 – AND– Zero Page
26 – ROL– Zero Page
27 – NOP
28 – PLP
29 – AND– Immediate
2A – ROL– Accumulator
2B – NOP
2C – BIT – Absolute
2D – AND– Absolute
2E – ROL– Absolute
2F – NOP
30 – BMI
31 – AND– (Indirect), Y
32 – NOP
33 – NOP
34 – NOP
35 – AND– Zero Page, X
36 – ROL– Zero Page, X
37 – NOP
38 → SEC
39 – AND– Absolute, Y
3A – NOP
3B – NOP
3C – NOP
3D – AND– Absolute, X
3E – ROL– Absolute, X
3F – NOP
40 – RTI
41 – EOR– (Indirect, X)
42 – NOP
43 – NOP
44 – NOP
45 – EOR– Zero Page
46 – LSR – Zero Page
47 – NOP
48 – PHA
49 – EOR– Zero Page
4A – LSR – Accumulator
4B – NOP
4C – JMP – Absolute
4D – EOR– Absolute
4E – LSR – Absolute
4F – NOP
50 – BVC
51 – EOR– (Indirect), Y
52 – NOP
53 – NOP
54 – NOP
55 – EOR– Zero Page, X
56 – LSR – Zero Page, X
57 – NOP
58 – CLI
59 – EOR– Absolute, Y
5A – NOP
5B – NOP
5C – NOP
5D – EOR– Absolute, ^
5E – LSR – Absolute, X
5F – NOP
60 – RTS
61 – ADC– (Indirect, X)
62 – NOP
63 – NOP
64 – NOP
65 – ADC– Zero Page
66 – ROR– Zero Page
67 – NOP
68 – PLA
69 – ADC– Immediate
6A – ROR– Accumulator
6B – NOP
6C – JMP – Indirect
6D – ADC– Absolute
6E – ROR– Absolute
6F – NOP
70 – BVS
71 – ADC– (Indirect), Y
72 – NOP
73 – NOP
74 – NOP
75 – ADC– Zero Page, X
76 – ROR– Zero Page, X
77 – NOP
78 – SEI
79 – ADC– Absolute, Y
7A – NOP
7B – NOP
7C – NOP
7D – ADC– Absolute, X NOP
7E – ROR– Absolute, X NOP
7F – NOP
80 – NOP
81 – STA – (Indirect, X)
82 – NOP
83 – NOP
84 – STY – Zero Page
85 – STA – Zero Page
86 – STX – Zero Page
87 – NOP
88 – DEY
89 – NOP
8A – TXA
8B – NOP
8C – STY – Absolute

8D – STA – Absolute
8E – STX – Absolute
8F – NOP
90 – BCC
91 – STA – (Indirect), Y
92 – NOP
93 – NOP
94 – STY – Zero Page, X
95 – STA – Zero Page, X
96 – STX – Zero Page, Y
97 – NOP
98 – TYA
99 – STA – Absolute, Y
9A – TXS
9B – NOP
9C – NOP
9D – STA – Absolute, X
9E – NOP
9F – NOP
A0 – LDY – Immediate
A1 – LDA – (Indirect, X)
A2 – LDX – Immediate
A3 – NOP
A4 – LDY – Zero Page
A5 – LDA – Zero Page
A6 – LDX – Zero Page
A7 – NOP
A8 – TAY
A9 – LDA – Immediate
AA – TAX
AB – NOP
AC – LDY – Absolute
AD – Absolute
AE – LDX – Absolute
AF – NOP
B0 – BCS
B1 – LDA – (Indirect), Y
B2 – NOP
B3 – NOP
B4 – LDY – Zero Page, X
B5 – LDA – Zero Page, X
B6 – LDX – Zero Page, Y
B7 – NOP
B8 – CLV
B9 – LDA – Absolute, Y
BA – TSX
BB – NOP
BC – LDY – Absolute, X
BD – LDA – Absolute, X
BE – LDX – Absolute, Y
BF – NOP
C0 – CPY – Immediate
C1 – CMP – (Indirect, X)
C2 – NOP
C3 – NOP
C4 – CPY – Zero Page
C5 – CMP – Zero Page
C6 – DEC – Zero Page
C7 – NOP
C8 – INY
C9 – CMP – Immediate
CA – DEX
CB – NOP
CC – CPY – Absolute
DC – CMP – Absolute
CE – DEC – Absolute
CF – NOP
D0 – BNE
D1 – CMP – (Indirect), Y
D2 – NOP
D3 – NOP
D4 – NOP
D5 – CMP – Zero Page, X
D6 – DEC – Zero Page, X
D7 – NOP
D8 – CLD
D9 – CMP – Absolute, Y
DA – NOP
DB – NOP
DC – NOP
DD – CMP – Absolute, X
DE – DEC – Absolute, X
DF – NOP
E0 – CPX – Immediate
E1 – SBC – (Indirect, X)
E2 – NOP
E3 – NOP
E4 – CPX – Zero Page
E5 – SBC – Zero Page
E6 – INC – Zero Page
E7 – NOP
E8 – INX
E9 – SBC – Immediate
EA – NOP
EB – NOP
EC – CPX – Absolute
ED – SBC – Absolute
EE – INC – Absolute
EF – NOP
F0 – BEQ
F1 – SBC – (Indirect), Y
F2 – NOP
F3 – NOP
F4 – NOP
F5 – SBC – Zero Page, X
F6 – INC – Zero Page, X
F7 – NOP
F8 – SED
F9 – SBC – Absolute, Y
FA – NOP
FB – NOP
FC – NOP
FD – SBC – Absolute, X
FE – INC – Absolute, X
FF – NOP

B BASIC Error Messages

As is commonly known among programmers, errors often occur in computer programs and must be corrected before the program can be properly executed. In practice the interruption of program execution can be avoided by using an error-handling routine with the help of the ONERR GOTO statement. The MPF-III has a set of error messages which will help you locate your errors. Therefore, this appendix contains an alphabetized list of some common error messages and their definitions frequently encountered by programmers.

The general format that BASIC uses to print the error messages depends on which execution mode is used to execute the instruction. In the immediate execution mode, an error message will appear as shown below:

```
?(name of error message) ERROR
```

In the deferred execution mode, an error message will appear with the error message as well as the line number so that you would be able to locate the error. Error messages of errors which occurred in the deferred execution mode will have the following format

```
?(name of error message) ERROR IN (line number in
                          which error occurred)
```

Listed below are some of the most frequently encountered messages and their definitions

ERROR MESSAGE	DEFINITION
BAD SUBSCRIPT	Attemppt to reference an array element which exceeds the dimension of the array. For example, when the dimension of an array are DIM A (2,2). Using LET A(1,1,1)=2 will cause an error.

CAN'T CONTINUE	Occurs when an attempt is made to execute a program which is : (1) a program in which an error has occurred and an attempt to resume execution was made, (2) a program to which a line of instruction has been deleted or added, (3) a program which does not exist. (4) a stop command executed or CTRL-C pressed, then some immediate execution result in error.
DIVISION BY ZERO	Any number divided by zero(Ø) is undefined and is considered to be an error.
FORMULA TOO COMPLEX	Occurs Upon execution of three or more IF...THEN statements having the same format.
ILLEGAL DIRECT	Using INPUT, DEF FN or DATA statement is not allowed in the immediate-execution mode.
ILLEGAL QUANTITY	An argument to be used in a math or string function was not in the specified range. This type of error may be encountered s a result of (a) an illegal argument was used with LEFT$, MID$, ON...GOTO, PEEK, POKE, RIGHT$, SPC, TAB, WAIT or any of the grahics functions. (b) a negative array subscript (c) a negative or zero argument is used with LOG. (d) a negative argument is used with SQR. (e) A^B where A is a negative number and B is a noninteger.

NEXT WITHOUT FOR	Means that the variable used in a NEXT statement and its corresponding FOR statement do not match or FOR was used without NEXT; and vice-versa.
OUT OF DATA	Attempt was made to execute a READ statement when all of the DATA statements in the program have already been used. Also can mean that too much data was read in by the READ statement or the data in the DATA sttements were insufficient.
OUT OF MEMORY	This error may occur due to the following : (1) FOR loops were more than 1Ø levels deep (2) GOSUB'S were more than 24 levels deep. (3) expressions used are too complex (4) parentheses were more than 36 levels deep (5) LOMEM: was set lower than the current value (6) HIMEM: was set too low (7) LOMEM: was set too high (8) variables in the program are too numerous to be contained in memory. (9) a program which is too large is to be stored in memory.
OVERFLOW	BASIC'S number format could not contain the final result of an arithmetical calculation because it may have been too large. If it should happen that the number is too small, then no error message is given and MPF-III will print a zero(Ø).

REDIM'D ARRAY	Another dimension statement having the same dimensions as another array used in the same program is encountered.
RETURN WITHOUT GOSUB	A RETURN statement was executed, but MPF-II could not find its GOSUB statement.
STRING TOO LONG	A string in your program is more than 255 characters long
SYNTAX ERROR	Is given when the syntax rules used in BASIC are not followed. Examples of this are : inserting a illegal character in an instruction or forgetting a parenthesis in an expression.
TYPE MISMATCH	Occurs when the variable name given in an assignment statement does not match with the argument given. For instance, this error will occur when a numeric variable name was given a string argument.
UNDEF'D STATEMENT	The line number specified in a GOTO, GOSUB or THEN does not exist in the program.
UNDEF'D FUNCTION	An attempt is made to reference a non-existent function which must have been defined by the user.

C MPF-III BASIC Reserved Words

Below is a list of the reserved words used in MPF-III. These words must not be used as variable names or as the first characters of the variable names.

&								
ABS	AND	ASC	AT	ATN				
BASS								
CALL	CHR$	CLEAR	COLOR=	CONT	COS			
DATA	DEF	DEL	DIM	DRAW				
EFFECT	END	EXP						
FN	FOR	FRE						
GET	GOSUB	GOTO	GR					
HCOLOR=	HGR	HGR2	HIMEM:	HLIN	HOME	HPLOT	HTAB	
IF	INPUT	INSTRU	INT	INVERSE	IN#			
LEFT$	LEN	LET	LIST	LOAD	LOG	LOMEM:		
MID$								
NEW	NEXT	NORMAL	NOT	NOTRACE				
ON	ONERR	OR						
PEEK	PLAY	PLOT	POKE	POP	POS	PRINT	PDL	PR#
READ	RECALL	REM	RESTORE	RESUME	RETURN	RIGHT#		
RND	ROT=	RUN						
SAVE	SCALE=	SCRN(	SGN	SHLOAD	SIN	SONG		
SPC(	SPEED=	SQR	STEP	STOP	STORE	STR$		
TAB	TAN	TEMPO	TEXT	THEN	TO	TRACE		
USR								
VAL	VLIN	VTAB						
WAIT								
XPLOT	XDRAW							

D ASCII Character Codes

In the binary key representation on the top of the following table, the three bits are used to show the status of ALT, the shift key (⇧), and the control key (CTRL). It is 1 if the corresponding key is pressed, otherwise it is Ø.

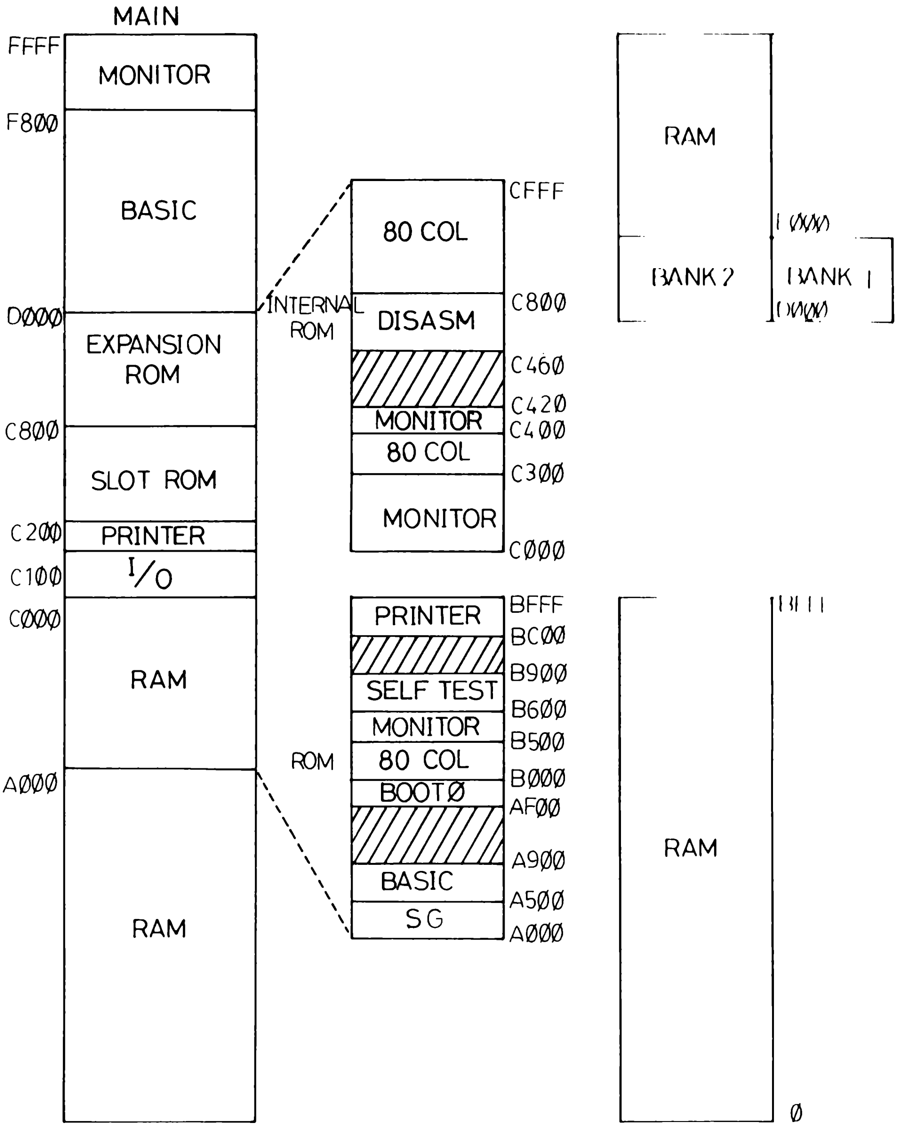
KEY	ALT/ ⇧ /CTRL							
	ØØØ	ØØ1	Ø1Ø	Ø11	1ØØ	1Ø1	11Ø	111
Ø INSC	ØØ	ØØ	BØ.	BØ	ØØ	ØØ	BØ	BØ
1 CLRL	Ø1	Ø1	B1	B1	Ø1	Ø1	B1	B1
2 ↓	8A	8A	B2	B2	8A	8A	B2	B2
3 CPES	02	02	B3	B3	02	02	B3	B3
4 ←	88	88	B4	B4	88	88	B4	B4
5	B5	B5	B5	B5	B5	B5	B5	B5
6 →	95	95	B6	B6	95	95	B6	B6
7 CLRS	03	03	B7	B7	03	03	B7	B7
8 ↑	8B	8b	B8	B8	8B	8B	B8	B8
9 HOME	04	04	B9	B9	04	04	B9	B9
. DELC	05	05	AE	AE	05	05	AE	AE
+ RUN	06	06	AB	AB	06	06	AB	AB
- LIST	1F	1F	AD	AD	1F	1F	AD	AD
HALT	93	93	93	93	93	93	93	93
BREAK	83	83	83	83	83	83	83	83

F1	20	2C	38	44	50	50	50	50
F2	21	2D	39	45	51	51	51	51
F3	22	2E	3A	46	52	52	52	52
F4	23	2F	3B	47	53	53	53	53
F5	24	30	3C	48	54	54	54	54
F6	25	31	3D	49	55	55	55	55
F7	26	32	3E	4A	56	56	56	56
F8	27	33	3F	4B	57	57	57	57
F9	28	34	40	4C	58	58	58	58
F10	29	35	41	4D	59	59	59	59
F11	2A	36	42	4E	5A	5A	5A	5A
F12	2B	37	43	4F	5B	5B	5B	5B

KEY	ALT/ ⇧ / CTRL							
	000	001	010	011	100	101	110	111
←	88	88	88	88	88	88	88	88
TAB	89	89	89	89	89	89	89	89
↵	8D	8D	8D	8D	8D	8D	8D	8D
ESC	9B	9B	9B	9B	9B	9B	9B	9B
SPACE	AO	AO	AO	AO	AO	AO	AO	AO
' "	A7	A7	A2	A2	A7	A7	A2	A2
, <	AC	AC	BC	BC	AC	AC	BC	BC
- _	AD	AD	DF	9F	AD	AD	DF	9F
. >	AE	AE	BE	BE	AE	AE	BE	BE
/ ?	AF	AF	BF	BF	AF	AF	BF	BF
0)	BO	Bo	A9	A9	5C	5C	5C	5C
1 !	B1	B1	A1	A1	5D	5D	5D	5D
2 @	B2	80	CO	80	5E	5E	5E	5E
3 #	B3	B3	A3	A3	5F	5F	5F	5F
4 $	B4	B4	A4	A4	60	60	60	60
5 %	B5	B5	A5	A5	7B	7B	7B	7B
6 ∧	B6	9E	DE	9E	7C	7C	7C	7C
7 &	B7	B7	A6	A6	7D	7D	7D	7D
8 *	B8	B8	AA	AA	7E	7E	7E	7E
9 (	B9	B9	A8	A8	7F	7F	7F	7F
: :	BB	BB	BA	BA	BB	BB	BA	BA
= +	BD	BD	AB	AB	BD	BD	AB	AB
[{	DB	9B	FB	9B	DB	9B	FB	9B
\ |	DC	9C	FC	9C	DC	9C	FC	9C
DEL	FF	FF	FF	FF	FF	FF	FF	FF

KEY	ALT/ ⇧ /CTRL							
	000	001	010	011	100	101	110	111
] }	DD	9D	FD	9D	DD	9D	FD	9D
` ~	EØ	EØ	FE	FE	EØ	EØ	FE	FE
A	E1	81	C1	81	61	61	61	61
B	E2	82	C2	82	62	62	62	62
C	E3	83	C3	83	63	63	63	63
D	E4	84	C4	84	64	64	64	64
E	E5	85	C5	85	65	65	65	65
F	E6	86	C6	86	66	66	66	66
G	E7	87	C7	87	67	67	67	67
H	E8	88	C8	88	68	68	68	68
I	E9	89	C9	89	69	69	69	69
J	EA	8A	CA	8A	6A	6A	6A	6A
K	EB	8B	CB	8B	6B	6B	6B	6B
L	EC	8C	CC	8C	6C	6C	6C	6C
M	ED	8D	CD	8D	6D	6D	6D	6D
N	EE	8E	CE	8E	6E	6E	6E	6E
O	EF	8F	CF	8F	6F	6F	6F	6F
P	FØ	9Ø	DØ	9Ø	70	70	70	70
Q	F1	91	D1	91	71	71	71	71
R	F2	92	D2	92	72	72	72	72
S	F3	93	D3	93	73	73	73	73
T	F4	94	D4	94	74	74	74	74
U	F5	95	D5	95	75	75	75	75
V	F6	96	D6	96	76	76	76	76
W	F7	97	D7	97	77	77	77	77
X	F8	98	D8	98	78	78	78	78
Y	F9	99	D9	99	79	79	79	79
Z	FA	9A	DA	9A	7A	7A	7A	7A

E MPF-III System Memory Map



F The RENUMBER Program

BASIC programs are executed in the order of line numbers. On some occasions, you may find it difficult to insert program lines due to the disorderly line-up of line numbers. The MPF-III provides the user with a RENUMBER program which is stored on the tape.

The RENUMBER program supports two variations for the user: RENUMBER and MERGE.

Follow the steps stated below:

1. Place the tape of the RENUMBER program in the cassette recorder and rewind the tape to the beginning.

2. Type LOAD

3. Press the PLAY button on the recorder. After two seconds, press the [↵] key on the keyboard.

4. When the screen displays "]●", type RUN to execute the RENUMBER program.

5. The following messages will be displayed. After you finished reading them, press the [↵] key.

```
****************************************
*                                      *
*          RENUMBER SERVICE            *
*                                      *
*                                      *
****************************************


RENUMBER   (DEFAULT VALUES)

& [First 10] [,Inc 10] [,Start 0] [,End 63999]


MERGE
```

```
&H   PUT PROGRAM ON HOLD
&M   MERGE TO PROGRAM ON HOLD



PRESS 'RETURN' TO CONTINUE...
```

Observe the following displayed line:

&[FIRST 10] [,INC 10] [, S0] [, E63999]

This is the format of the RENUMBER command with the default values.

1) &: The identifier of the RENUMBER command, indicating the following characters and numbers are used for the RENUMBER command.

2) First X: X is used to specify the new line number of the first program line.

3) Inc X: X is used to specify the increment.

4) Start X: X is used to indicate the starting line number from which the RENUMBER will be executed.

5) End X: X is used to indicate the line number at which renumbering will end.

MERGE: To merge two programs into one.

1) &H: Hold a program.

2) &M: Merge the held program with another.

6. Press the [↵] key and you will see the following messages. Now you can input program lines and use RENUMBER and MERGE.

```
  RENUMBER IS INSTALLED AND READY
IF YOU USE 'FP', 'HIMEM', OR 'MAXFILES'

 YOU WILL HAVE TO RE-RUN RENUMBER
```

Examples:

Key in the following programs. Observe carefully how the RENUMBER commands work in these examples.

```
10 INPUT A,B
20 IF A < = B THEN GOTO 50
30 PRINT "A IS LARGER"
40 GOTO 10
50 IF A < B THEN GOTO 80
60 PRINT "THEY ARE THE SAME"
70 GOTO 10
80 PRINT "B IS LARGER"
90 GOTO 10
```

A.

```
]&F5,I5

]LIST

5 INPUT A,B
10 IF A < = B THEN GOTO 25
15 PRINT "A IS LARGER"
20 GOTO 5
25 IF A < B THEN GOTO 40
30 PRINT "THEY ARE THE SAME"
35 GOTO 5
40 PRINT "B IS LARGER"
45 GOTO 5
```

B.

```
]&F32, S20,I10

]LIST

5 INPUT A,B
10 IF A < = B THEN GOTO 42
15 PRINT "A IS LARGER"
32 GOTO 5
42 IF A < B THEN GOTO 72
52 PRINT "THEY ARE THE SAME"
62 GOTO 5
72 PRINT "B IS LARGER"
82 GOTO 5
```

C.

```
]&F40, S42, E72,I5
```

```
]LIST

5 INPUT A,B
10 IF A < = B THEN GOTO 40
15 PRINT "A IS LARGER"
32 GOTO 5
40 IF A < B THEN GOTO 55
45 PRINT "THEY ARE THE SAME"
50 GOTO 5
55 PRINT "B IS LARGER"
82 GOTO 5
```

NOTE: In the preceding examples, we have used the abbreviated forms as shown below:

F=First
S=Start
E=End
I=Inc

MERGE

Key in Program 1 as follows:

```
]10 FOR A=65 TO 90
]20 PRINT CHR$(A)
]30 NEXT A
```

Type &H to hold this program and key in Program 2:

```
]&H
PROGRAM ON HOLD, USE "&M" TO RECOVER

]50 FOR A = 48 TO 57
]60 PRINT CHR$(BA);
]70 NEXT A
```

Type &M to merge these two programs

```
]&M

]LIST

10 FOR A = 65 TO 90
20 PRINT CHR$(A);
30 NEXT A
50 FOR A = 48 TO 57
60 PRINT CHR$(A);
70 NEXT A
```

```
]RUN
ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789
]
```

To merge two programs which are read from magnetic types or diskettes, follow the steps stated below:

1) Type LOAD filename 1
2) Type &H
3) Type LOAD filename 2
4) Type &M

Note: When you merge two programs, the program lines with the same line numbers will both be listed. In this case, there might be unpredictable results during program execution. Therefore, it is recommended that you enumber the program lines before you merge two programs.

G More on the Function Keys

On the MPF-III keyboard, you may find 12 function keys (F1-F12). Using these keys in combination with the SHIFT, CTRL, and ALT keys, 60 codes can be generated as described in 1.2.3.

If MPF-III DOS is not used, the 12 function keys can all be defined by the user.

When MPF-III DOS is used, press ALT and the function keys simultaneously will generate the commands as follows:

```
ALT F1: LOAD          ALT F7: BLOAD
    F2: SAVE              F8: BSAVE
    F3: OPEN              F9: BRUN
    F4: CLOSE             F10: POSITION
    F5: READ              F11: APPEND
    F6: WRITE             F12: CATALOG
```

You can define the function keys according to the ASCII character codes in Appendix D.

For example:

```
                F1 → 20
          SHIFT F1 → 38
           CTRL F2 → 2D
    SHIFT  CTRL F2 → 45
```

In the example that follows, we will define F1-F8 as the eight SG rhythms and F9-F12 as the four SG Instruments.

```
F1: Waltz        F5: Blues        F9: Piano
F2: Rumba        F6: Rock         F10: Bell
F3: Tango        F7: Swing        F11: Xylophone
F4: March        F8: Chacha       F12: Organ
```

After you set up the music score for a song, you may press the function key to select the instrument and the rhythm.

Try the following example:

Music score SSS (6ØE6-61E7)

```
*60E0.6167

60E0- 15 73 31 77 FF FF 00 73
60E8- 15 73 31 73 33 73 35 75
60F0- 35 73 36 73 35 75 34 73
60F8- 33 73 35 75 35 75 00 73
6100- 15 73 31 73 33 73 35 75
6108- 35 73 36 73 35 75 34 73
6110- 33 73 34 77 00 73 15 73
6118- 17 73 32 73 34 75 34 73
6120- 35 73 34 75 33 73 32 73
6128- 34 75 34 75 00 75 35 73
6130- 34 73 33 73 00 73 32 73
6138- 31 73 32 73 00 73 31 73
6140- 17 73 31 78 00 75 16 73
6148- 31 73 31 73 32 73 31 76
6150- 16 73 15 75 33 77 00 75
6158- 15 73 32 73 32 73 33 73
6160- 32 76 31 73 33 78 00 75
```

```
*6168.61E9

6168- 16 73 31 73 31 73 32 73
6170- 31 76 16 73 15 75 33 77
6178- 00 75 31 73 31 73 31 73
6180- 32 73 33 75 32 73 31 73
6188- 32 75 00 75 00 73 15 73
6190- 31 73 33 73 35 75 35 73
6198- 36 73 35 75 34 73 33 73
61A0- 35 75 35 75 00 75 15 73
61A8- 31 73 33 73 35 75 35 73
61B0- 36 73 35 75 34 73 33 73
61B8- 34 77 00 73 15 73 17 73
61C0- 32 73 34 75 34 73 35 73
61C8- 34 75 33 73 32 73 34 75
61D0- 34 75 00 75 35 73 34 73
61D8- 33 75 32 73 31 73 32 75
61E0- 31 73 17 73 31 78 FF FF
61E8- 32 05
```

```
]LIST

10  TEXT : HOME
401 D$ =  CHR$ (4): PRINT D$;"BLOAD SSS"
410  HOME
411  VTAB 1: HTAB 5: PRINT "**** MUSIC WITH BASS PLAY
    ****";
412  VTAB 5: HTAB 15: PRINT "F9 PIANO";
413  VTAB 7: HTAB 15: PRINT "F10 BELL";
414  VTAB 9: HTAB 15: PRINT "F11 xylophone";
415  VTAB 11: HTAB 15: PRINT "F12 ORGAN": VTAB 22: HTAB
     8: PRINT "(MUSICAL INSTRUMENT FUNCTION)";
416  VTAB 24: HTAB 10: PRINT "ENTER YOUR SELECTION ";:
     GET AN$
417  IF AN$ =  CHR$ (40) THEN EE = 1: GOTO 425
418  IF AN$ =  CHR$ (41) THEN EE = 2: GOTO 425
419  IF AN$ =  CHR$ (42) THEN EE = 3: GOTO 425
420  IF AN$ =  CHR$ (43) THEN EE = 4: GOTO 425
423  GOTO 416
425  HOME
430  VTAB 1: HTAB 5: PRINT "**** MUSIC WITH BASS PLAY
     ****";
440  VTAB 3: HTAB 15: PRINT "F1 WALTZ";
450  VTAB 5: HTAB 15: PRINT "F2 RUMBA";
470  VTAB 7: HTAB 15: PRINT "F3 TANGO";
480  VTAB 9: HTAB 15: PRINT "F4 MARCH";
510  VTAB 11: HTAB 15: PRINT "F5 BLUES";
511  VTAB 13: HTAB 15: PRINT "F6 ROCK";
512  VTAB 15: HTAB 15: PRINT "F7 SWING";
513  VTAB 17: HTAB 15: PRINT "F8 CHA-CHA";
515  VTAB 19: HTAB 15: PRINT "0  MAIN-MENU";
516  VTAB 22: HTAB 9: PRINT "(MUSIC BASS FUNCTION)";
520  VTAB 24: HTAB 10: PRINT "ENTER YOUR SELECTION ";:
     GET AN$
530  IF AN$ =  CHR$ (32) THEN FF = 1: GOTO 2050
540  IF AN$ =  CHR$ (33) THEN FF = 2: GOTO 2050
550  IF AN$ =  CHR$ (34) THEN FF = 3: GOTO 2050
560  IF AN$ =  CHR$ (35) THEN FF = 4: GOTO 2050
570  IF AN$ =  CHR$ (36) THEN FF = 5: GOTO 2050
580  IF AN$ =  CHR$ (37) THEN FF = 6: GOTO 2050
590  IF AN$ =  CHR$ (38) THEN FF = 7: GOTO 2050
600  IF AN$ =  CHR$ (39) THEN FF = 8: GOTO 2050
610  IF AN$ = "0" THEN 410
615  GOTO 520
1554  PLAY 6 * 4096 + 14 * 16 + 6,FF: RETURN
2050  INSTR EE: TEMPO 5: GOSUB 1554: GOTO 410
```

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